

# Growth needs more than access

Three rigorous economics papers · two current applied-development studies and one agricultural-economics classic. The following article pages are edited reading excerpts, not complete replacements for the originals.

## LITERATURE PULSE

The three papers ask why a seemingly valuable input may produce little until complementary capacity is in place. In Ethiopia, a lower-cost app reaches entrepreneurs initially but loses them before they finish training; neither delivery mode raises firm performance in the short run. Across ASEAN, simulated trade openness yields larger gains when domestic productivity rises too, though these are model scenarios rather than observed policy effects. The classic developing-country study links farm inputs and cereal yields to later growth and labor movement out of agriculture, using fertilizer geography as an instrument. **Literature trend:** the return to access depends on completion, productive capability, and how gains move through the wider economy. The evidence spans randomized treatment effects, structural simulations, and historical IV estimates; their different assumptions should not be conflated.

01 CURRENT / GENERAL INTEREST · STARTS ON PAGE 2 · World Bank Economic Review · 2026 · peer reviewed · open WPS 10992 edition, CC BY 3.0 IGO

## Testing the Promise of Digital Scaling: In-Person versus App-Based Training for Women Entrepreneurs

Can a cheap mobile app deliver business training as effectively as a classroom? An Addis Ababa randomized trial assigned 2,000 smartphone-owning women entrepreneurs to app-based training, the same in-person curriculum, or control. App completion was 22%, versus 71% in person, despite similar initial take-up; neither mode meaningfully improved business performance at three to six months. The study cleanly isolates delivery mode, but its connected, educated sample, short follow-up, and possible measurement limits constrain generalization. The surprise is that lower delivery cost did not translate into effective scale when sustained attention was scarce.

02 CURRENT / GROWTH AND TRADE · STARTS ON PAGE 17 · World Bank Policy Research Working Paper 11440 · September 2026 · CC BY 3.0 IGO

## Openness and Productivity Reform: Lessons for Middle-Income Economies from Thailand and Other ASEAN

Why do trade reforms pay off unevenly across ASEAN? A calibrated quantitative trade model joins input-output linkages and labor frictions, then simulates lower trade costs and higher productivity using international benchmarks. For Thailand, the model implies roughly 41% higher real income under the trade-cost scenario and 55% under the productivity scenario; the combined scenario is larger than either alone. These are model-based counterfactuals, not forecasts or causally observed reform effects. Their scale depends on assumed feasible shocks and parameters. The useful idea is complementarity: access to markets matters more when domestic firms can exploit it.

03 CLASSIC / AGRICULTURAL TRANSFORMATION · STARTS ON PAGE 32 · Journal of Development Economics 127 · 2017 · classic · CC BY 4.0 PMC text edition

## Fertilizing Growth: Agricultural Inputs and Their Effects in Economic Development

Can better farm inputs trigger economy-wide development? Using a developing-country panel from 1961–2001, the authors relate fertilizer, improved seeds, and water to cereal yields, then instrument fertilizer access with global price changes interacted with transport distance to fertilizer production. Their estimates imply a half-ton yield gain is associated with 14–19% higher GDP per person and a 4.6–5.6-point lower agricultural labor share five years later. Exclusion of other distance-by-price channels is contestable, and the historical macro sample limits present-day prediction. It remains a vivid test of whether farm productivity can power structural change.

*Editorial note: references and appendices were omitted where possible; no included article contributes more than 15 text pages. Full citations and source links remain in the catalog. AI-written synopses were checked against the included source pages.*

# Testing the Promise of Digital Scaling: In-Person versus App-Based Training for Women Entrepreneurs

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# 1 Introduction

Business training has long been a staple tool in development policy, aimed at alleviating human capital constraints among entrepreneurs. Annually, at least USD 1 billion is allocated to such programs in low- and middle-income countries, the vast majority of which are delivered in person (McKenzie and Woodruff, 2014). However, research indicates that the outcomes of these programs are modest at best (McKenzie et al., 2023), with even smaller or negligible benefits for women. These limited benefits are often due to restrictive social norms that impose mobility and time constraints on women (Jayachandran, 2021; Ubfal, 2024).

Following extensive experimentation with digital learning during the COVID-19 pandemic, policy makers have increasingly recognized the potential of digital technologies to deliver training at scale. Digital delivery can increase cost-effectiveness and scalability, by substantially reducing expenses related to facility rentals, supplies, and staff (Chang, 2016). Furthermore, digital training offers full flexibility in terms of location and time — potentially addressing constraints that women face, such as unpaid care work and limited mobility (Bandiera and Zipfel, 2019; Beegle and Rubiano-Matulevich, 2020). However, access to digital training may require cell phone or computer ownership, internet connectivity, and a basic level of digital skills. Furthermore, virtual training is not necessarily more cost-effective than traditional methods, particularly when trainers still need to be recruited and some content is delivered live (Davies et al., 2024). Additionally, digital applications often experience high dropout rates due to insufficient instructor and peer engagement (Bawa, 2016). Therefore, it is crucial to address the evidence gap regarding the impact of digital training, and to evaluate empirically the trade-offs between digital and in-person modalities.

We conduct a randomized controlled trial (RCT) to compare the impact of digital versus in-person business training, using the same curriculum with the same target population. Specifically, we compare in-person delivery against delivery via a specialized ed-tech mobile app, adapted to the local context. We deliberately select a sample predisposed to engage with digital training and likely to benefit from it, aiming to maximize the chances of demonstrating the feasibility and impact of app-based learning.

Our study specifically targets established, growth-oriented women entrepreneurs. We focus on this sample due to the relatively weak evidence supporting the impact of training for individuals — particularly women — who have not yet become entrepreneurs or are subsistence entrepreneurs (McKenzie et al., 2023). The training employs a curriculum that is more advanced than traditional, basic business training, and is designed to benefit entrepreneurs who have already acquired foundational business skills. Our study thus also provides evidence on whether tailoring training content to above-subsistence women entrepreneurs may

produce a greater impact than traditional business training, even if delivered in person.

Despite our targeted approach —and high initial take-up rates for both the app (80%) and the in-person training (76%) — our results reveal very low completion rates for the digital training (22%), compared to high completion rates for the in-person training (71%). Low usage of the app persists despite additional efforts to incentivize engagement, including weekly reminders, cash-prize draws, and making the app available offline.

Both treatment arms produce statistically significant but modest improvements in business knowledge, with no larger gains observed for the in-person training despite its higher completion rate. Six months after accessing the training, both treatments yield negligible impacts on business practices and performance. Taken at face value, our results suggest that even more advanced business training targeted at existing, growth-oriented entrepreneurs may not work. Our data do not allow us to distinguish whether the lack of impact is due to lack of capacity to implement learning, or because other constraints to growth are binding.

The details of our experiment are as follows. We recruit a sample of 2,000 women in Addis Ababa, Ethiopia, who own or manage a business and are looking to expand it. The recruited businesses are small but above subsistence levels, with 79% having at least one employee and a median of three employees. We further focus on women with at least secondary education, given the target level of the curriculum. Our eligibility criteria include access to a smartphone and the internet, as well as a reported interest in digital training modalities.

The app-based training program offers interactive business education accessible on both iOS and Android devices, and is available in Amharic and English. It consists of eleven modules derived from mobile MBA preparatory courses provided by a U.S. ed-tech firm. These modules encompass a wide range of topics, including entrepreneurship, marketing, accounting, and finance. Originally designed for pre-MBA preparatory courses in the United States, the materials were meticulously adapted to suit the Ethiopian context in collaboration with the ed-tech firm. Additionally, we partnered with a local business training provider to develop an in-person course that mirrors the app-based training. This in-person course follows the same syllabus and is delivered in eleven two-hour sessions, conducted twice a week over a six-week period.

As an additional screening device and to increase the likelihood of training participation, we invited women to an in-person information session. During this session, women received detailed information about the training content and its two delivery modalities. At the conclusion of the session, women were randomly assigned through a public lottery to one of three groups: the app-based training, the in-person training, or a control group. The control group was justified due to the over-subscription to the app during its pilot phase and the limited availability of in-person training slots. The app-based group was offered assistance to

download the app, log in and start using it. To understand the importance of this assistance, we conduct an additional experiment with a similar sample, in which we offer access to the app *without previously inviting women to information sessions*. We observe minimal take-up of even one app module. The information sessions may also have contributed to the relatively high take-up of the in-person training — although we do not test this experimentally.

Recognizing the need for sustained incentives to ensure engagement with the app-based learning and to replicate the group interaction inherent in the in-person training, we introduced an additional treatment arm. In this arm, half of the women assigned to the app-based training were also included in social messaging groups and received weekly prompts to discuss the course content with their peers. However, this intervention did not impact participation or outcomes. We observed negligible engagement in the groups, partly due to intermittent government blocks on the popular social messaging app used, and partly due to a lack of interest among women in virtual group interactions. Consequently, when reporting results, we pool all women assigned to the app-based training, regardless of their inclusion in the social messaging groups.

In exploratory analysis, self-reported digital skills at baseline emerges as the only significant predictor of completing the app-based course. This finding aligns with the observation that, despite screening for smartphone ownership and interest in digital training, some women reported difficulties in using the app and troubleshooting basic issues. Such observations indicated lower digital literacy and digital confidence among this population than initially expected. Internet connectivity did not seem to hinder participation, as most of the sample reported having moderately to very reliable internet access. Furthermore, as previously mentioned, the app was available for offline use in the later phases of the experiment. Our findings indicate that neither internet connectivity nor the app’s offline functionality were correlated with take-up.

In our follow-up survey, qualitative responses from women also frequently cite “not having the time” as a reason for not completing the app modules. This response suggests that the commitment device of scheduled in-person training and the presence of an in-person peer group may play a crucial role in fostering training completion. Commitment could be important for addressing self-control issues, negotiating with employers for time off from work, or coordinating with family members to take time off from unpaid care work. We do not find a correlation between app take-up and proxies for “other-control” issues, such as hours spent working per day, hours spent on care work, marital status, or having children under the age of five. However, we cannot rule out self-control issues.

To our knowledge, ours is the first experiment to compare the effects of app-based training versus in-person training with identical content, conducted in parallel for the same target

## 2.3 In-person business training

The content for the in-person training mirrors that of the app-based training (Appendix Table A14), but is adapted for a face-to-face format. This training is delivered by a local provider with proven expertise in business training. While the primary teaching method is slideshow presentations, the classes also incorporate interactive techniques such as role plays, plenary discussions, brainstorming activities, and group projects to maintain participant engagement. To optimize the learning experience, participants in this group completed a needs assessment survey, indicating their preferences for session schedules, venues, and childcare facilities. Based on these preferences, they were organized into cohorts of 25, ensuring their individual needs were met. The training course was structured to cover all modules in individual sessions over a six-week period, with two-hour sessions held twice a week. These sessions were staggered so that only five training groups were active at any given time. Similar to the app-based training, participants who attend at least eight sessions receive a completion certificate.

## 3 Experimental Design

### 3.1 Recruitment, information sessions, and randomization

Our sampling frame is derived from the Ethiopia Women Entrepreneurship Development Project (WEDP), which has registered over 40,000 women business owners in its Management Information System, with approximately 17,700 based in Addis Ababa. WEDP is a national program run by the Government of Ethiopia, with financial and technical assistance from the World Bank. The project targets growth-oriented women-owned enterprises, offering interventions that include information, training, and financial support.

To build our sampling frame, we reached out to all 17,700 entrepreneurs and successfully conducted screening phone interviews with approximately 6,000 women. The phone interview assessed participants based on the following eligibility criteria: i) access to and knowledge of how to use a smartphone; ii) access to the internet; iii) completion of at least a high school education; iv) ability to understand either English or Amharic; v) being a business owner or active manager of an operating business; and vi) the business having 30 or fewer employees. This screening process resulted in about 4,000 eligible women.

Respondents who met the eligibility criteria were asked at the end of the screening survey if they were interested in a free-of-cost business training mobile app. If they answered positively, they were then invited to attend in-person information sessions about the training,

and offered a travel compensation to do so. These sessions took place monthly in four central locations over a span of four months, from March to June 2023. Recruitment was staggered to allow the in-person training sessions to run with five simultaneous classes of 25 participants, according to the training provider’s capacity. Overall, 2,000 participants attended information sessions.

Our eligibility criteria naturally resulted in a specialized sample being invited to the information sessions. However, attendance at these sessions further filtered the participant pool. Appendix Table A1 compares the characteristics of eligible participants who attended the information sessions with those who did not, using data from the screening phone interview. On average, attendees of the information sessions are older, more educated, own larger businesses, live closer to the session venues, and are less likely to prefer only the app-based training over in-person training. The greater distance from information venues and the higher preference for app-based training among non-attendees may suggest that they face more significant mobility constraints or belong to a slightly younger generation that favors digital learning over in-person training.

During the information sessions, participants were asked to complete the baseline survey on their smartphones under the supervision of enumerators. The survey took approximately an hour and included questions about the respondent and their household characteristics, their primary and secondary business characteristics and performance, business practices, access to finance, networks, and their digital skills. After completing the survey, participants were provided with details on the range of interventions being offered and the randomization process. Participants were then invited individually to draw a token from an urn containing an equal number of four different tokens corresponding to three treatment groups and one control group. The three treatment arms included app-based training alone, app-based training combined with social networking, and in-person training. Given that the total number of participants at each session was not usually divisible by four, this in-person lottery generated some small, statistically insignificant differences in the number of participants assigned to each treatment arm, with 495, 502, and 520 women assigned to the first, second, and third treatment arms, respectively, while 483 women were assigned to the control group.

Women assigned to the app-based training were given immediate and unrestricted access to the app, along with support for the download and login process, and a tutorial on its features. Those assigned to the in-person training were informed that their sessions would commence within the following weeks, with a training duration of approximately six weeks. They were also told that they would be contacted by the trainers to receive more details on the exact timing and location of the sessions.

As discussed below, conducting in-person information sessions increases the overall cost

of offering the course. However, without in-person help to download the app and start using it, we believed that very few women would experiment with the app. To test this assumption, we conducted an additional experiment with 228 women who had not attended information sessions but expressed interest in the app. We shared with them the login credentials to access the app via phone call and text message. A random half of the sample received three basic help and troubleshooting calls from agents. Overall, only about 10 percent of this sample, —regardless of phone call assistance — completed at least one module, and just three percent completed at least eight modules. This additional experiment confirms the need for an initial in-person orientation to generate take-up for app-based training. The information sessions may also have acted as an effective screening device for the in-person training, and contributed to the relatively high take-up of the in-person training, although we do not test this experimentally. For comparison, [Alibhai et al. \(2019\)](#) find take-up of just 41% for a personal initiative training offered to a similar population of women entrepreneurs in Addis Ababa.

## 3.2 Sample characteristics

Table 1 describes the main characteristics of the sample and checks for balance across the app-based training, in-person training, and control arms. As mentioned above, take-up for the networking intervention is very low, and we observe no significant differences at baseline or in outcomes between those assigned to the app-based training with versus without the networking arm. Therefore, in all tables, we present pooled results for the two app-based training treatment arms. Results that include the three treatment arms are reported in the Online Appendix (Tables A2 - A5).

The average participant is a 38-year-old mother, often the head of her household. She manages a business with an average of four employees (median of three employees) and reports implementing around five out of eight recommended business practices.<sup>2</sup> The majority of participants have completed more than secondary education and operate businesses in the service (41 percent) or trade (32 percent) sectors. The average monthly household income is USD 881 (at nominal exchange rate), monthly business revenues are USD 1,311, and monthly profits are USD 268.

Table 1 shows that the randomization resulted in only one imbalanced variable (respondent’s main business is over five years old) out of twenty between the in-person training and

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<sup>2</sup>Of the eight questions, three focus on marketing and product innovation, two address supplier competitiveness, and three pertain to forecasting and accounting. Although more than half of these practices were implemented at baseline, marketing practices were adopted to a lesser extent compared to the other two categories. The curriculum extensively covered marketing and accounting, but it did not emphasize supplier competitiveness as much.

control groups. Overall, two variables out of twenty show significant imbalances between the app-based training and the control groups at the five percent level, contributing to a rejection of the F-test of joint orthogonality across these groups. Specifically, the app-based group is younger and significantly more likely to have completed post-secondary education. All of our regressions use post-double LASSO to select the control variables to include, often picking up the imbalanced variables. Additionally, we find that our results are robust when using a matching estimator with exact matching on a post-secondary education dummy (Appendix Table A8).

### 3.3 Follow-up data collection

Over the duration of the interventions, the research team gathered administrative data pertinent to both training modalities. For the app-based training, the team utilized an administrative portal to monitor course progression in real time. For the in-person training, attendance was documented by the training facilitators.

From September to December 2023, approximately six months after participants first engaged with the training, we conducted a short follow-up telephone survey. This survey was a streamlined version of the baseline survey and included additional queries related to business performance, business practices, and the motivations behind participants' level of involvement in the training or absence thereof. We successfully re-interviewed 1,928 women during this follow-up, resulting in a low attrition rate of only five percent from the original cohort. Importantly, the rate of attrition and the profile of those who dropped out did not significantly vary across the different treatment groups (Appendix Table A9).

## 4 Results

### 4.1 Empirical strategy

We estimate the effects of being assigned to the app-based or in-person training using the following ANCOVA specification:

$$Y_{i1} = \beta_0 + \beta_1 T_{1i} + \beta_2 T_{2i} + \beta_3 Y_{i0} + \beta_4 X'_{i0} + \beta_5 S_i + \epsilon_{i1}, \quad (1)$$

where  $Y_{i1}$  is the outcome variable for individual  $i$  measured at follow-up ( $t = 1$ ).  $T_{1i}$ , and  $T_{2i}$  are dummy variables taking the value of one if the individual is assigned to the app-based or the in-person training, respectively.  $Y_{i0}$  is the baseline value of the outcome variable.  $X'_{i0}$  is a vector of LASSO-selected baseline control variables from Table 1, which includes

imbalanced variables such as age and education.  $S_i$  is a vector of randomization strata, containing dummies for each of the information sessions where public lotteries occurred. We report Eicker-White standard errors that are robust to heteroskedasticity. The parameters  $\beta_1$  and  $\beta_2$  correspond to the intention-to-treat effect of being assigned to the app-based or in-person training compared to the control group, respectively.

## 4.2 Treatment compliance and training take-up

Table 2 presents the effects of being invited to the training on training take-up, compared to women in the control group. According to the administrative app data, no woman in the control group or the in-person training group had access to the app-based training. Similarly, no woman assigned to the app-based training was invited to the in-person training. There was only one instance where a woman in the control group was mistakenly invited to the in-person training.

Column 1 indicates that in both the app-based and in-person training groups, over 75% of individuals completed at least one module or session, respectively. For the app-based training, this initial engagement was partly facilitated by enumerators who encouraged and guided participants through the first part of the first module during the information sessions. However, when it comes to course completion—defined as completing at least eight modules of the app-based training or attending at least eight sessions of the in-person training, which is required to obtain the certificate—there is a significant difference between the two treatment modalities. Column 2 reveals that only 22% of the app-based group completed at least eight modules, whereas 71% of the in-person group attended at least eight training sessions.

Columns 3-5 of Table 2 provide a detailed analysis of the initiation and completion rates of the courses. On average, participants in the app-based training started 5.2 modules (measured by simply opening the module), partially completed 3.7 modules (measured by completing at least half of the lessons in a module), and fully completed 3.3 modules (measured by passing a short quiz at the end of the module). In contrast, participants in the in-person training attended an average of 7.2 sessions. Even under the conservative assumption that the average in-person participant only paid attention to half of the content—suggesting that in-person attendance is best compared to partial completion of an app module—these figures indicate that in-person participants engaged with twice as much of the course content as app-based users (attending 7.2 sessions versus partially completing 3.7 app modules).

Column 6 presents self-reported completion data, which largely corroborates the administrative data for in-person training. However, it shows that app-based participants report completion rates more in line with the initiation of modules (Column 3) rather than their

served minimal interaction. Additionally, we sent weekly reminders to encourage participants to complete the app’s modules. To further incentivize module completion, we introduced small cash-prize lotteries, which were announced via the instant messaging service and text messages. Although we observed some increases in module completion a few days following these announcements (see Appendix Figure A3), the overall completion rate of the app-based training remained low, despite the various incentives and nudges provided.

### 4.3 Impacts on primary and secondary outcomes

Table 3 Panel A presents results from ANCOVA regressions for our pre-specified primary outcomes: business survival and performance.<sup>3</sup> The results indicate no significant effects of being assigned to either modality of business training on these outcomes.

Panel A, Column 1 shows the effect on whether the business remained open at follow-up. Approximately 91% of businesses in both the treatment and control groups remained open, with almost no difference across treatment arms. Columns 2-6 indicate no significant treatment effects on business sales and profits. The effects on winsorized sales (Column 5) and profits (Column 3) for the main business are imprecisely estimated, with point estimates very close to zero. However, the estimated coefficients and standard errors do not rule out positive effects of up to 15 percent. This pattern holds true when considering both the main and secondary businesses operated by women (Columns 4 and 6, respectively). The pre-specified profits and sales index, which combines outcomes from Columns 3-6, shows more precisely estimated null effects, allowing us to rule out effects above 0.1 standard deviations.

Panel B in Table 3 presents results for our pre-specified secondary outcomes: business inputs, capital, practices, and knowledge. Once again, we observe almost no statistically significant impact from either of the two treatments.

We first verify that business owners gained knowledge from both modalities of training. There is a statistically significant, albeit small, impact of two percentage points for both treatment arms on business knowledge, as measured by a seven-question test (Column 9). Notably, the impact on business knowledge does not differ across treatment arms, despite a much higher completion rate for the in-person training. This effect represents only a 5 percent increase over the control mean of 43 percent correct answers. A similar effect size is reported by Davies et al. (2024) for synchronous online training.

We find precisely estimated null effects on business practices, measured by a subset of eight business practices from McKenzie and Woodruff (2017). The control group reports implementing 62 percent of these practices, and the share is similar for the two training

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<sup>3</sup>Our pre-analysis plan was registered at: <https://www.socialscienceregistry.org/trials/11193>.

groups (Column 8). There are also no significant treatment effects on a capital index that combines business assets and investments (Columns 5-7), nor on an inputs index that includes the number of employees, hours worked, and hours the business is open (Columns 1-4). For the inputs index, the number of hours the business is open, and the number of hours worked ( $p = 0.12$ ), we observe statistically significant differences between the two treatment arms. Specifically, there are positive point estimates for the in-person training and negative point estimates for the app-based training, although these differences are not statistically significant at the 5 percent level. Overall, the lack of effects on inputs and investment, as well as the lack of changes in business practices, is consistent with the null effects observed for business performance.

#### 4.4 Robustness and treatment effect heterogeneity

Appendix Tables A2 - A7 examine the robustness of the results by excluding controls and by using three instead of two treatment arms, as pre-specified. We confirm the lack of any statistically significant effect on both primary and secondary outcomes.

In addition, Appendix Tables A10 - A13 show no statistically significant heterogeneity by pre-specified variables, including education, baseline profits, digital skills, and business sector. We also explore heterogeneous effects using the generic machine learning approach of Chernozhukov et al. (2018), and find no sub-groups with differential impacts.

Finally, Figures A1 - A2 show that for both the in-person and app-based training, quantile treatment effects for the level of profits are below the average treatment effect for all but the top percentile, for which effects are very imprecisely estimated. For business practices, we observe some statistically significant effects in the middle of the distribution, with lower effects in the tails.

### 5 Costs of app-based versus in-person training

We consider it informative as part of the “proof-of-concept” aspect of the study to quantify the potential cost savings from using app-based delivery versus a face-to-face modality. Such evidence on costs could be useful for future interventions, especially if relatively cost-neutral changes to the content of similar training programs are found to lead to stronger effects.

We calculate an average cost per person assigned to receive the training of USD 278 for the app-based training and of USD 511 for the in-person training. These costs are based on the detailed financial proposals to train 997 individuals via the app, and 521 in person, hence we note that the comparison is unfavorable to the in-person training given the

presence of fixed costs and economies of scale. The app-based training costs can be broken down into fixed costs for developing the curriculum, adapting and translating the material, plus app programming, and staff support for app maintenance (82%). Variable costs include recruitment (16%) and user licenses (2%). For the in-person training, the average cost includes fixed costs for developing the curriculum, staff costs, and adapting and translating the material (65%). In our setting, most of these costs were absorbed by the app developers, but we allocate them evenly between the two training types as they draw from the same curriculum. Variable costs are training implementation (27%), and recruitment through in-person information sessions (8%). The fact that the average per-potential-participant cost of app-based training is 54% of that of in-person training indicates that while cost savings from app-based training are significant, they are not as substantial as might be anticipated. This finding aligns with that of [Davies et al. \(2024\)](#), who identified high costs for synchronous digital training, which includes significant costs for online live training. The average cost per participant *who completed the course* is mechanically higher for both training modalities, and significantly more so for the app-based training due to lower completion rates. However, the key advantage of asynchronous training is that it can be offered to additional participants at a low *marginal* cost. Specifically, the marginal cost for our app-based training, based on the variable costs reported above, was one-third of that for in-person training: USD 49 for the app-based training versus USD 178 for in-person training. Most of the variable costs for the app-based training are attributed not to licenses or other costs directly linked to the app, but rather to recruitment, particularly the in-person information sessions. These recruitment costs are often excluded from cost calculations in other studies but are typically crucial for training providers to ensure a minimum threshold number of participants. In the case of our app-based training, these in-person sessions also assisted participants in downloading and learning how to navigate the app. Our additional experiment showed that the in-person sessions were essential to obtain any engagement with the app. For in-person training, in addition to recruitment costs, the main variable expenses were for venues and trainers.<sup>4</sup> We further note that these marginal costs may be non-linear when scaling. For example, marginal costs of recruitment may increase if it is increasingly difficult to find eligible participants ([Maffioli et al., 2023](#)), or decrease if the app gains viral popularity.

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<sup>4</sup>For comparison, the average per-participant cost of an in-person 5-7 day training course reported in [Van Lieshout and Mehtha \(2017\)](#) is USD 177; while marginal costs are not reported. Our costs are higher compared to shorter training programs in Ethiopia implemented via the government Technical and Vocational Education and Training (TVET) structure, for example, the training in [Alibhai et al. \(2019\)](#) with a cost of USD 30.

children. However, we cannot rule out self-control issues.

Future work might consider targeting app-based training at an even more selected group, such as “Gen-Z” entrepreneurs who are highly familiar with using similar apps. It might also be fruitful to test an app that is more gamified, which has been shown to produce success in other contexts (Melo et al., 2023; Lafortune et al., 2024). Gamification might keep digitally literate users more engaged and even encourage less digitally literate users to learn skills such as troubleshooting through engagement with the app. On a broader level, policy makers should prioritize initiatives to improve digital literacy. Additionally, other access and adoption challenges should be addressed, such as providing affordable hardware and enhancing mobile connectivity infrastructure.

That said, the lack of impacts on business practices and outcomes, even from the well-attended in-person training, suggests that a fundamental issue may lie in the content or in barriers to implementing the learning. We cannot rule out the possibility that women did improve their business practices or outcomes following the in-person training, but that our survey measures were not well-calibrated to capture these impacts within a short time frame. However, taken at face value, our results suggest that even more sophisticated and targeted business training may not produce significant impacts, even when the modality ensures high completion rates.

An open question is whether women realized during the in-person training that it would not produce the desired impact, and if so, why they continued to attend. It is possible that the social benefits of attending or the peer pressure costs of not attending outweighed the disutility and opportunity cost of participating in a training they did not expect to improve their business. Either way, the results suggest that in-person training may offer a stronger commitment device than app-based training. If effective training programs are designed, the challenge for digital training will be to find alternative commitment devices to encourage course completion.

Table 2: Impacts on Training Completion

|                         | Admin Data                                       |                                                |                                                             |                                                                         |                                                               | Self-reported                                                 |
|-------------------------|--------------------------------------------------|------------------------------------------------|-------------------------------------------------------------|-------------------------------------------------------------------------|---------------------------------------------------------------|---------------------------------------------------------------|
|                         | (1)<br>Partial compliance<br>(at least 1 course) | (2)<br>Full compliance<br>(at least 8 courses) | (3)<br>App modules started<br>or Training sessions attended | (4)<br>App modules partially completed<br>or Training sessions attended | (5)<br>App modules completed<br>or Training sessions attended | (6)<br>App modules completed<br>or Training sessions attended |
| App-based training (T1) | 0.80***<br>(0.01)                                | 0.22***<br>(0.01)                              | 5.21***<br>(0.14)                                           | 3.72***<br>(0.14)                                                       | 3.33***<br>(0.13)                                             | 6.85***<br>(0.13)                                             |
| In-person training (T2) | 0.76***<br>(0.02)                                | 0.71***<br>(0.02)                              | 7.21***<br>(0.19)                                           | 7.21***<br>(0.19)                                                       | 7.21***<br>(0.19)                                             | 8.07***<br>(0.21)                                             |
| Observations            | 2000                                             | 2000                                           | 2000                                                        | 2000                                                                    | 2000                                                          | 1928                                                          |
| Control mean            | 0.00                                             | 0.00                                           | 0.00                                                        | 0.00                                                                    | 0.00                                                          | 0.00                                                          |
| P-val t1=t2             | 0.09                                             | 0.01                                           | 0.01                                                        | 0.01                                                                    | 0.01                                                          | 0.01                                                          |

Notes: Outcomes used in Columns 1-5 are constructed from administrative data, whereas the outcome used in Column 6 is constructed from self-reported data. Outcome 1 takes the value of 1 if the respondent completed at least 1 module on the app or attended 1 session of the in-person training course. Outcome 2 takes the value of 1 if respondent completed at least 8 modules on the app or attended at least 8 sessions of the in-person training course. Outcome 3 indicates the number of modules started on the app, or the number of classes attended, as there is no similar measure for in-person training. Outcome 4 indicates the number of modules partially completed, where the respondent completes at least half the lessons in a module, or the number of classes attended, as there is no partial measure for in-person attendance. Outcome 5 indicates the number of modules completed on the app, or the number of in-person sessions attended. Outcome 6 indicates the number of courses completed on the app, or the number of in-person sessions attended according to self-reported data from the follow-up survey. Results are obtained from OLS regressions of the outcome on two treatment dummies, controlling for LASSO selected controls, and randomization cohort fixed effects. Robust standard errors are in parentheses. \* $p < 0.1$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$

Table 3: Impacts on Primary and Secondary outcomes

|                                       | Panel A: Impact on Primary outcomes                     |                                    |                                               |                                                             |                                                 |                                                               |                                   |                                         |                                 |
|---------------------------------------|---------------------------------------------------------|------------------------------------|-----------------------------------------------|-------------------------------------------------------------|-------------------------------------------------|---------------------------------------------------------------|-----------------------------------|-----------------------------------------|---------------------------------|
|                                       | (1)                                                     | (2)                                | (3)                                           | (4)                                                         | (5)                                             | (6)                                                           |                                   |                                         |                                 |
|                                       | Business was operating<br>at follow-up<br>(Yes 1; No 0) | Profits and sales<br>Index<br>(SD) | Last month's profit<br>Main business<br>(USD) | Last month's profit<br>Main and secondary business<br>(USD) | Last month's revenues<br>Main business<br>(USD) | Last month's revenues<br>Main and secondary business<br>(USD) |                                   |                                         |                                 |
| App-based training (T1)               | -0.02<br>(0.01)                                         | 0.01<br>(0.04)                     | 7.32<br>(23.08)                               | -5.03<br>(24.27)                                            | 3.17<br>(91.42)                                 | -32.16<br>(96.50)                                             |                                   |                                         |                                 |
| In-person training (T2)               | 0.01<br>(0.02)                                          | 0.01<br>(0.05)                     | -1.00<br>(26.72)                              | -15.26<br>(27.78)                                           | -24.59<br>(105.17)                              | -69.60<br>(111.89)                                            |                                   |                                         |                                 |
| Observations                          | 1928                                                    | 1928                               | 1928                                          | 1928                                                        | 1928                                            | 1928                                                          |                                   |                                         |                                 |
| Control mean                          | 0.91                                                    | 0.00                               | 398.85                                        | 421.81                                                      | 1420.53                                         | 1513.11                                                       |                                   |                                         |                                 |
| P-val t1=t2                           | 0.09                                                    | 0.89                               | 0.73                                          | 0.69                                                        | 0.77                                            | 0.70                                                          |                                   |                                         |                                 |
| Panel B: Impact on Secondary outcomes |                                                         |                                    |                                               |                                                             |                                                 |                                                               |                                   |                                         |                                 |
|                                       | (1)                                                     | (2)                                | (3)                                           | (4)                                                         | (5)                                             | (6)                                                           | (7)                               | (8)                                     | (9)                             |
|                                       | Inputs<br>Index<br>(SD)                                 | Number of<br>employees             | Number of<br>hours worked                     | Number of<br>hours open                                     | Capital<br>Index<br>(SD)                        | Business assets<br>(USD)                                      | Made investments<br>(Yes 1; No 0) | Business practice score<br>(Proportion) | Knowledge score<br>(Proportion) |
| App-based training (T1)               | -0.03<br>(0.03)                                         | 0.08<br>(0.16)                     | -0.10<br>(0.18)                               | -0.32*<br>(0.18)                                            | 0.01<br>(0.04)                                  | 1450.08<br>(1516.67)                                          | 0.01<br>(0.02)                    | 0.00<br>(0.01)                          | 0.02**<br>(0.01)                |
| In-person training (T2)               | 0.04<br>(0.04)                                          | 0.25<br>(0.19)                     | 0.17<br>(0.20)                                | 0.03<br>(0.20)                                              | 0.01<br>(0.04)                                  | 1209.45<br>(1701.71)                                          | 0.00<br>(0.02)                    | 0.02<br>(0.01)                          | 0.02*<br>(0.01)                 |
| Observations                          | 1928                                                    | 1928                               | 1928                                          | 1928                                                        | 1928                                            | 1928                                                          | 1928                              | 1928                                    | 1928                            |
| Control mean                          | 0.00                                                    | 3.42                               | 7.69                                          | 9.63                                                        | 0.00                                            | 16035.48                                                      | 0.09                              | 0.62                                    | 0.43                            |
| P-val t1=t2                           | 0.03                                                    | 0.31                               | 0.12                                          | 0.04                                                        | 0.94                                            | 0.88                                                          | 0.75                              | 0.23                                    | 0.79                            |

Notes: Panel A presents impacts on primary outcomes and Panel B presents impacts on secondary outcomes. Panel A, Outcome 1 is a binary variable indicating if the business was open at follow-up. Outcome 2 shows a profits and sales index, which is constructed by averaging the z scores (relative to the control group) of main and secondary business profits and sales, both IHS transformed and levels. Outcome 3 shows winsorized last month profits for the main business in USD. Outcome 4 shows last month winsorized profits for both the main and secondary business in USD. Outcome 5 shows last month winsorized sales for the main business in USD. Outcome 6 shows last month winsorized sales for both the main and secondary business in USD. Panel B, Outcome 1 is an inputs index, which is constructed by averaging the z scores (relative to the control group) of outcomes in Columns 2-4. Outcome 2 is the winsorized number of employees in the main business. Outcome 3 is the number of hours worked by the entrepreneur in her main business. Outcome 4 shows the number of hours the main business is open in a day. Outcome 5 is a capital index, which is constructed by averaging the z scores (relative to the control group) of assets and investments, both IHS transformed and in levels. Outcome 6 shows the winsorized value of business assets in USD. Outcome 7 is a binary variable which takes the value of 1 if the entrepreneur made a large investment in the past 6 months. Outcome 8 is the proportion of business practices implemented in the past 6 months. Outcome 9 is the proportion of correct answers to business knowledge questions. Results are obtained from OLS regressions of the outcome on two treatment dummies, controlling for the baseline value of the outcome, LASSO selected controls, and randomization cohort fixed effects. Robust standard errors are in parentheses. \* $p < 0.1$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$

# Openness and Productivity Reform: Lessons for Middle-Income Economies from Thailand and other ASEAN\*

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# 1 Motivation and Introduction

This paper studies how middle-income economies can convert trade openness and productivity-enhancing reforms into higher real income and welfare. Many countries have used trade integration to accelerate development, but the transition from middle-income to high-income status has proven more difficult. Openness alone does not guarantee domestic value creation, technological upgrading, or broad-based welfare gains. At the same time, productivity reforms may have limited impact if firms face high costs in accessing imported inputs, foreign markets, logistics networks, standards systems, and global production platforms. The central question of this paper is therefore not whether countries should prioritize trade openness or productivity reform, but under what conditions the two reinforce each other.

The Association of Southeast Asian Nations (ASEAN) provides a useful setting for studying this question because it is both highly integrated and highly heterogeneous. The region includes economies deeply embedded in Asian manufacturing value chains, economies specialized in commodities and resource-based activities, economies that have pursued export-led industrialization, and economies with more selective or cautious approaches to trade integration. This heterogeneity allows us to examine when openness generates large welfare gains, when productivity improvements matter more, and when the two channels are complementary. It also allows us to study how differences in production structure, input-output linkages, and global value chain participation shape the returns to reform.

The question is especially relevant in the current global environment. Globalization is not disappearing, but it is becoming more fragmented, more regionalized, and more rule-intensive. Trade remains resilient in aggregate, but firms increasingly operate in a world of policy uncertainty, regulatory divergence, standards requirements, certification procedures, rules of origin, and logistics constraints. These frictions affect not only whether goods cross borders, but whether firms can connect to, operate within, and upgrade inside production networks.

We refer to this broader policy environment as one of qualified openness. Qualified openness means openness supported by the institutions, infrastructure, and capabilities that

to foreign markets. Broader trade-cost reductions capture the combined effect of lowering the frictions that shape the movement of goods and inputs across borders. Productivity improvements capture firm capabilities, technology adoption, managerial quality, innovation, and production efficiency. The combined scenario captures the policy-relevant case in which trade-cost reductions and productivity improvements occur together.

The main result is that trade-cost reductions and productivity improvements are complements, but the strength of each channel differs across ASEAN economies. Lower trade frictions expand access to inputs and markets, but firms can exploit these opportunities more effectively when they have the capabilities to scale, comply, and innovate. Conversely, productivity improvements generate larger gains when firms can access imported inputs and foreign markets at lower effective cost. This complementarity is central to the development challenge facing middle-income economies: openness alone does not guarantee upgrading, and capability upgrading is constrained when firms face high interoperability costs.

The paper contributes to the policy discussion in three ways. First, it reframes the middle-income trade challenge as one of qualified openness rather than simply more openness. Second, it quantifies the aggregate gains from reducing effective trade costs and improving productivity in a unified general equilibrium framework. Third, it shows that these gains are heterogeneous across ASEAN economies, reflecting differences in production structure, GVC participation, sectoral linkages, and domestic capabilities. The implication is that trade policy, industrial policy, and regulatory capability should be treated as a connected reform agenda aimed at turning openness into welfare gains and domestic value creation.

The rest of the paper is organized as follows. Section 2 documents stylized facts on import, export, and domestic constraints in selected ASEAN economies, using Thailand as a detailed illustrative case where the available evidence is richest. Section 3 presents the model. Section 4 describes the data and quantification. Section 5 reports the counterfactual results for ASEAN economies. Section 6 discusses policy implications. Section 7 concludes.

## 2 Stylized Facts: Import, Export, and Domestic Constraints in ASEAN

This section documents the empirical motivation for the model counterfactuals. The paper distinguishes among import-cost reductions, export-cost reductions, broader trade-cost reductions, and productivity improvements. These channels are not abstract modeling categories. They correspond to observable constraints facing Thailand and other ASEAN economies: the cost of sourcing imported intermediates, the cost of qualifying for foreign markets, the domestic frictions that affect logistics and regulatory compliance, and the firm-level capabilities needed to turn trade participation into higher domestic value added.

The evidence points to three stylized facts. First, ASEAN economies are highly heterogeneous in their mode of integration. Some economies, such as Singapore, Viet Nam, Thailand, and Malaysia, are deeply embedded in manufacturing value chains, while others, such as Indonesia, have lower measured GVC participation and a production structure less centered on cross-border manufacturing networks. Second, in manufacturing-oriented ASEAN economies, import access is a central component of export competitiveness because production relies heavily on foreign intermediates. Third, the most binding trade frictions are increasingly related to standards, certification, logistics, customs, documentation, and regulatory recognition rather than tariffs alone. These facts motivate the paper’s focus on qualified openness: openness supported by the institutions and capabilities that allow firms to connect, comply, and upgrade within production networks.

### 2.1 ASEAN heterogeneity and the structure of GVC participation

ASEAN provides a useful setting because its economies participate in trade in different ways. Table 1 reports GVC participation for selected ASEAN economies and comparators. Three patterns are worth emphasizing. First, Thailand is highly integrated into GVCs: GVC-related activity accounts for 28.8 percent of output overall and 46.9 percent of manufacturing output. Second, Thailand’s manufacturing integration is asymmetric. Backward participation in manufacturing is 10.6 percent, while forward participation is

3.4 percent, indicating strong reliance on imported intermediates and a more limited role in upstream value creation. Third, this pattern is not unique to Thailand. Viet Nam and Singapore also exhibit high backward participation in manufacturing, whereas GVC participation in Indonesia and the Philippines is substantially lower.

Table 1: GVC participation and asymmetry in selected economies

| Country     | Overall     |            |            | Manufacturing |             |            |
|-------------|-------------|------------|------------|---------------|-------------|------------|
|             | GVC         | Backward   | Forward    | GVC           | Backward    | Forward    |
| Indonesia   | 9.0         | 2.2        | 2.6        | 15.0          | 3.7         | 2.7        |
| Malaysia    | 25.7        | 4.3        | 4.7        | 36.9          | 5.9         | 3.6        |
| Philippines | 12.1        | 3.9        | 2.4        | 18.1          | 5.4         | 1.9        |
| Singapore   | 44.8        | 9.6        | 5.2        | 53.1          | 16.1        | 5.0        |
| Thailand    | <b>28.8</b> | <b>6.8</b> | <b>3.8</b> | <b>46.9</b>   | <b>10.6</b> | <b>3.4</b> |
| Viet Nam    | 34.0        | 9.8        | 3.6        | 49.4          | 15.0        | 2.9        |
| World       | 13.0        | 2.6        | 2.6        | 23.7          | 5.1         | 3.0        |

*Notes:* Values refer to 2024 and are percent shares of output. Backward participation reflects the use of foreign inputs in domestic production. Forward participation reflects domestic value added embodied in other countries' exports. Source: WITS GVC Module, output tables (WITS, 2026), based on Borin et al. (2025). The output tables are based on Asian Development Bank (ADB) Multi-Regional Input-Output Database.

This structure has direct implications for the model. In economies with high backward participation, import-cost reductions can generate large welfare effects because cheaper imported intermediates reduce downstream production costs. Conversely, high backward participation also creates exposure to supply disruptions, customs delays, certification bottlenecks, and regulatory changes affecting imported inputs. For Thailand, Viet Nam, Singapore, and Malaysia, import access is therefore not only a consumption margin; it is a production and competitiveness margin.

The contrast with Indonesia highlights the importance of heterogeneity. Indonesia's overall and manufacturing GVC participation are lower than Thailand's, Viet Nam's, Malaysia's, and Singapore's. This suggests that the gains from trade-cost reductions may be smaller

only as tariff liberalization. They include regulatory recognition, harmonized standards, mutual recognition of testing and certification, improved rules-of-origin administration, and lower fixed costs of entering demanding foreign markets. This interpretation is especially relevant for countries seeking to move beyond labor-intensive production or basic assembly into higher-value manufacturing and embedded services.

## **2.4 Domestic constraints: Policy frictions, firm capabilities, and services embedded in manufacturing**

The third constraint is domestic. Trade reforms generate larger gains when firms have the capabilities to exploit them. Conversely, productivity improvements generate larger gains when firms can access inputs and markets at low effective cost. Domestic constraints therefore include both policy frictions and capability gaps.

Thailand's domestic trade regime illustrates the interaction between these two margins. Its simple average MFN tariff is approximately 9.9 percent, with a sharp difference between agriculture, where the average is approximately 28.3 percent, and non-agriculture, where it is approximately 7.0 percent. This means that Thailand's domestic regime still contains tariff dispersion, especially in agriculture. However, for the manufacturing upgrading agenda emphasized in this paper, the more binding constraints are non-tariff and implementation-related: SPS and TBT measures, licensing, certification delays, limited recognition, and regulatory predictability.

This creates a two-worlds trade policy environment. Thai exporters operate in foreign markets where tariffs are often low and compliance is the primary margin of competition, while the domestic regime still contains tariff and non-tariff barriers that can raise costs for firms. The implication is that competitiveness depends increasingly on regulatory capability, not tariff protection. Domestic institutions must help firms meet foreign requirements, certify products, manage documentation, and operate reliably across regulatory systems.

Domestic capability gaps are also visible in technology and services. Thailand has strengths in applied and system-level technologies, especially in activities linked to electric vehicles, electronics, machinery, power electronics, connectors, thermal management, and control

inputs and broader market access. Countries with weaker GVC participation or more commodity-oriented structures may benefit relatively more from productivity improvements and domestic capability upgrading. Across all cases, however, the key policy-relevant mechanism is complementarity: lower trade costs raise the returns to domestic capabilities, while stronger domestic capabilities raise the returns to openness.

The central implication is that ASEAN’s middle-income development challenge is not simply to increase trade volumes. It is to reduce the import, export, and domestic constraints that prevent firms from using trade relationships productively. For Thailand, the evidence is especially clear: the country is already integrated into global and regional production networks, but its ability to upgrade depends on reducing qualification frictions, improving regulatory and logistics performance, and strengthening the domestic capabilities that allow firms to capture more value within those networks.

### 3 Model

In this section, we introduce a general equilibrium trade model based on Eaton and Kortum, 2002 with additional components to improve its fit to data and policy relevance. Similar to them, we consider a system with many locations and differentiated varieties, where consumers can source each variety from a different location based on prices and transportation costs. Prices are directly impacted by the underlying productivity distribution in locations. Different from them, we allow multiple sectors which are continua of differentiated goods, and we incorporate input output linkages into the production functions. These two modifications make the framework similar to the seminal model introduced by Caliendo and Parro, 2015. In addition to these two significant differences, we assume that workers have differentiated productivity across sectors and they face a static discrete choice problem to choose the optimal sector of employment. This is a simplified version of the Artuc et al., 2010 model. The simplification facilitates calibration of the system for all countries in the sample. See Redding, 2016, Galle et al., 2022, Lee, 2020, Artuc and Sommer, 2024.

Consider  $N$  countries, indexed with either  $m$  or  $n$ , where agents derive utility from consuming varieties  $\omega_j$  categorized into  $J$  sectors. The varieties are combined by agents in country  $n$  to

produce sectoral composites:

$$Q_j^n \equiv \left[ \int (Q^n(\omega_j))^{\frac{\sigma-1}{\sigma}} d\omega_j \right]^{\frac{\sigma}{\sigma-1}},$$

where  $Q^n(\omega_j)$  is the quantity of variety  $\omega_j$  used, and  $\sigma$  is the elasticity parameter. The sectoral composites enter the utility of country  $n$  agents as

$$U^n = \prod_j (Q_j^n)^{\gamma_j^n},$$

where  $Q_j^n$  is the consumed quantity of sector  $j$  composite, and  $\gamma_j^n$  is the Cobb-Douglas share parameter.

The production process requires combining intermediate inputs with labor:

$$q^m(\omega_j) = A_j^m z^m(\omega_j) \left[ \prod_k (Q_k^m(\omega_j))^{\beta_{jk}^m} \right]^{\alpha_{jM}^m} [\widetilde{L}^m(\omega_j)]^{\alpha_{jL}^m},$$

where  $Q_k^m(\omega_j)$  is the sector  $k$  composite and  $\widetilde{L}^m(\omega_j)$  is the effective labor units used in  $\omega_j$  production. The effective unit of labor is different from headcount and this point will be clarified later in the text. Cobb-Douglas production function shares and material input shares add up to one,  $\alpha_{jL}^m + \alpha_{jM}^m = 1$  and  $\sum_k \beta_{jk}^m = 1$ . The technology parameter  $A_j^m$  is country-sector specific and reflects the overall productivity. As usual, assume that the country-variety-specific productivity variable  $z^m(\omega_j)$  is distributed Frechet with shape  $\theta$  and scale  $[\Gamma(1 - \frac{1}{\theta})]^{-1}$ , where  $\theta > 1$ . The cost of producing one unit of  $\omega_j$  is equal to

$$\tilde{c}^m(\omega_j) = \varrho_j^m \frac{(c_{jM}^m)^{\alpha_{jM}^m} (w_j^m)^{\alpha_{jL}^m}}{A_j^m z^m(\omega_j)},$$

where  $\varrho_j^m = (\alpha_{jM}^m)^{-\alpha_{jM}^m} (\alpha_{jL}^m)^{-\alpha_{jL}^m}$  is the constant cost multiplier which does not play any role in the solution,  $c_{jM}^m$  is the price of intermediate input and  $w_j^m$  is the sector-specific wage. Consumers can source varieties locally if they are produced locally or import them from abroad. Destination country  $n$  consumers must pay  $d_j^{nm}$  times the price when they purchase

$$L_j^m = \left( \frac{w_j^m}{W^m} \right)^\nu L^m. \quad (7)$$

The headcount of workers in sector  $j$  is different from the effective units of labor input. The latter is given as  $\widetilde{L}_j^m = (w_j^m/W^m)^{\nu-1} L^m$ . Since there is only one factor of production (excluding the input materials), the total income is simply equal to  $W^m L^m$ .

Labor market clearing under perfect competition implies that the value of wage bill implied by firms' optimization problem should be equal to the total income of workers. In this case, Cobb-Douglas technology simply implies a constant share of the wage bill in total output, when the production is optimized.

Consider sectoral output,

$$Y_j^m = \sum_n X_j^{nm}. \quad (8)$$

where  $X_j^{nm}$  is the value of sector  $j$  flows from  $m$  to  $n$ . Then, the labor market clearing condition discussed above implies that

$$w_j^m \left( \frac{w_j^m}{W^m} \right)^{\nu-1} L^m = \alpha_j^m Y_j^m, \quad (9)$$

which simply forces the total wage income of all sector  $j$  workers characterized by the labor allocation problem to be equal to the wage bill paid by producers.

**Market clearing.** Total nominal income is equal to

$$I^n = W^n L^n, \quad (10)$$

which is the sum of disposable income,  $W^n L^n$ . The expenditure is

$$E^n = I^n - S^n, \quad (11)$$

where  $S^n$  is the savings, which is assumed to be fixed.

Then, goods market clearing implies that exports from  $m$  to  $n$ , defined as  $X_j^{nm}$ , is equal to

ICIO gives us the trade array ( $X_j^{nm}$ ), share of labor in production ( $\alpha_{jL}^m$ ), and share of materials in production ( $\alpha_{jM}^m$ ). From these three key arrays, we can construct most of the remaining variables and parameters required to solve the model using exact hat algebra. These variables and parameters are: sectoral output, initial total income, initial total expenditure, initial savings rate, initial import shares, initial export shares, initial demand for consumption goods and intermediate inputs. ICIO 2025 covers 1995-2022, with 81 countries and 50 industries. Many of the large non-ASEAN economies are included in the dataset such as China, USA, EU, Japan, Brazil, India, UK, Türkiye, South Africa, Mexico, Saudi Arabia and Korea, along with Brunei Darussalam, Cambodia, Indonesia, Lao PDR, Malaysia, Myanmar, Philippines, Singapore, Thailand, and Viet Nam from ASEAN. We aggregate the industries into 21 groups with and work with the following sectors: 1. Agriculture, hunting, forestry, fishing; 2. Mining and quarrying, energy producing products; 3. Mining and quarrying, non-energy producing products; 4. Food products, beverages and tobacco; 5. Textiles, textile products, leather and footwear; 6. Wood and paper; 7. Coke and refined petroleum products; 8. Chemical and chemical products; 9. Pharmaceuticals, medicinal chemical and botanical products; 10. Rubber and plastics products; 11. Other non-metallic mineral products; 12. Basic iron and steel; 13. Basic non-iron metals; 14. Fabricated metal products; 15. Computer, electronic and optical equipment; 16. Electrical equipment; 17. Machinery and equipment; 18. Motor vehicles, trailers and semi-trailers; 19. Other transport equipment; 20. Other manufacturing; 21. All Services.

PWT provides the change in country-specific productivity over time ( $A_j^m$ ), which is assumed to be identical across sectors due to data limitations. These series allow us to implement counterfactual evolution of productivity series over time.

The remaining two parameters, trade elasticity  $\theta$  and labor elasticity  $\nu$  are taken from the literature. The trade elasticity is assumed to be  $\theta = 4.0$  based on Simonovska and Waugh, 2014. We take labor elasticity equal to  $\nu = 1.44$ , following Arias et al., 2018.

**Counterfactual trade cost and productivity changes.** We calculate the implied trade

costs using the standard Head and Ries, 2001 formula

$$d_j^{nm} = \left( \frac{X_j^{nn} X_j^{mm}}{X_j^{nm} X_j^{mn}} \right)^{\frac{1}{2\theta}},$$

using the ICIO trade data as described earlier. Then, we calculate the change in trade costs as  $\widehat{d_j^{nm}} = d_{j,2022}^{nm}/d_{j,1995}^{nm}$ . In the counterfactual simulations, we will impose the change in the trade costs faced by the countries that switch to high income status to ASEAN country trade costs. The change in productivity is also calculated similarly as  $\widehat{A_j^m} = A_{j,2022}^m/A_{j,1995}^m$  using PWT data.

## 5 Results: Trade Costs, Productivity, and the Returns to Qualified Openness

This section uses the model to quantify how changes in effective trade costs and productivity affect real income. The counterfactuals are designed to isolate five channels. The first reduces import costs, capturing lower costs of sourcing foreign intermediates through improvements in logistics, customs, regulatory recognition, connectivity, and non-tariff barriers. In particular, we calculate the average change in import costs experienced by countries that switch to high-income status. Then impose this number to ASEAN countries, instead of using their own import cost changes. The formula is provided in the previous section. The second reduces export costs similarly again using the formula provided in the previous section, capturing improved access to foreign markets and smoother cross-border procedures for selling abroad. The third reduces broader trade costs, combining the main channels through which goods and inputs move across borders. The fourth increases total factor productivity (TFP), capturing improvements in firm capabilities, technology adoption, managerial practices, supplier quality, and production efficiency. Similar to trade cost shocks, we calculate the average productivity increases for countries that switch to high-income status and apply them to ASEAN countries instead of using their own observed productivity increases. The fifth combines trade-cost reductions with productivity improvements, combining all counterfactual shocks.

Table 6 reports the percentage change in real income under these benchmark shocks for Thailand and selected ASEAN comparators. The results should be interpreted as counterfactual changes relative to the baseline equilibrium, not as unconditional forecasts. They show the model-implied returns to reducing effective trade costs and raising productivity under a common benchmark. The benchmark shocks are based on trade-cost reductions and productivity gains observed among economies that reached high-income status since 1995.<sup>2</sup>

Table 6: Percentage change in real income after benchmark shocks

| Country     | Import cost  | Export cost  | Trade cost   | TFP          | Trade cost + TFP |
|-------------|--------------|--------------|--------------|--------------|------------------|
| Viet Nam    | 27.50        | 20.75        | 72.01        | 105.21       | 215.04           |
| Malaysia    | 18.09        | 15.45        | 46.61        | 58.09        | 127.53           |
| Cambodia    | 34.52        | 13.72        | 58.47        | 38.40        | 116.78           |
| Thailand    | <b>16.72</b> | <b>12.51</b> | <b>41.17</b> | <b>54.78</b> | <b>114.15</b>    |
| Philippines | 8.20         | 7.01         | 19.67        | 45.48        | 72.26            |
| Lao PDR     | 12.29        | 6.08         | 20.79        | 44.93        | 73.54            |
| Myanmar     | 8.64         | 4.30         | 16.98        | 59.56        | 82.78            |
| Indonesia   | 6.41         | 2.92         | 12.20        | 46.58        | 62.82            |

*Notes:* Values are percentage changes in real income relative to the baseline equilibrium. The import-cost scenario captures reductions in the effective cost of sourcing foreign inputs. The export-cost scenario captures reductions in the effective cost of selling abroad. The trade-cost scenario captures broader reductions in effective trade costs. The TFP scenario captures productivity improvements. The combined scenario applies trade-cost reductions together with productivity improvements.

Several results stand out. First, the returns to reform differ substantially across ASEAN economies. Viet Nam, Malaysia, Cambodia, and Thailand show especially large gains from trade-cost reductions, while Indonesia, Myanmar, Lao PDR, and the Philippines show smaller gains from trade-cost reductions but still sizable gains from productivity improvements. This heterogeneity is central to the paper’s argument. Middle-income

<sup>2</sup>The benchmark is calibrated using economies that transitioned to high-income status since 1995 that also appear in the ICIO dataset. The 18 economies are: Bulgaria; Chile; Costa Rica; Croatia; Czechia; Estonia; Greece; Hungary; Korea, Rep.; Latvia; Lithuania; Malta; Poland; Romania; Russian Federation; Saudi Arabia; Slovak Republic; and Slovenia.

economies do not benefit from openness in the same way. The welfare effects depend on production structure, GVC participation, reliance on imported inputs, sectoral linkages, and the ability of firms to transform market access into domestic value creation.

Second, import-cost reductions generate larger welfare gains than export-cost reductions in all countries reported in the table. For Thailand, a reduction in import costs increases real income by 16.72 percent, compared with 12.51 percent for a reduction in export costs. This asymmetry is consistent with the importance of imported intermediates in production networks. Lower import costs reduce the price of composite intermediates, improve downstream competitiveness, and propagate through the input-output structure of the economy. The result suggests that market access should not be understood only as access to foreign consumers. For GVC-oriented economies, access to foreign inputs is also a core component of competitiveness.

Third, broader reductions in trade costs generate substantially larger gains than either the import or export margin alone. The total impact is also larger than the sum of the two shocks when applied separately, indicating significant complementarities. For Thailand, the trade-cost scenario increases real income by 41.17 percent. For Viet Nam, Malaysia, and Cambodia, the corresponding gains are 72.01 percent, 46.61 percent, and 58.47 percent, respectively. These effects reflect the general equilibrium nature of the model. Lower trade frictions reduce consumer price indices, lower intermediate input costs, change sourcing shares, raise output in competitive sectors, and affect wages through labor reallocation. The impact is therefore not limited to the direct cost reduction at the border; it propagates through production networks and domestic factor markets.

Fourth, productivity improvements generate large gains across the region. For Thailand, the TFP scenario increases real income by 54.78 percent. The corresponding gains are 105.21 percent for Viet Nam, 58.09 percent for Malaysia, 45.48 percent for the Philippines, 44.93 percent for Lao PDR, 59.56 percent for Myanmar, and 46.58 percent for Indonesia. These results highlight that openness is not enough. The ability of firms and sectors to benefit from trade depends on their productivity, managerial capacity, technology adoption, supplier quality, and ability to meet market and regulatory requirements. Productivity improvements

reduce unit costs and allow firms to exploit larger input and output markets more effectively. Fifth, the combined trade-cost and productivity scenario produces the largest gains in every country. For Thailand, real income rises by 114.15 percent when trade-cost reductions and productivity improvements occur together. This exceeds the sum of the separate trade-cost and productivity effects, which indicates complementarity between the two channels. The same pattern holds across the ASEAN comparators. Lower trade frictions raise the returns to productivity improvements by expanding access to inputs and markets. Productivity improvements raise the returns to trade-cost reductions by allowing firms to scale, comply, and compete. The model therefore supports a joint reform strategy rather than a sequencing in which trade openness and capability upgrading are treated as separate policy agendas.

Thailand provides a useful illustration of the broader middle-income challenge. Its gains are substantial, but not the largest in the region. Thailand benefits strongly from lower trade costs because of its deep integration in manufacturing GVCs and reliance on imported intermediates. It also benefits strongly from productivity improvements because upgrading requires stronger domestic capabilities, supplier quality, and movement into higher-value functions. The combined scenario shows that Thailand's welfare gains are not driven by market access or productivity alone, but by the interaction between the two.

The results also clarify the role of interoperability. In the model, reductions in iceberg trade costs are represented as lower  $d_j^{nm}$ . In the policy interpretation, these reductions should be understood broadly. They include not only lower tariffs or transport costs, but also faster customs clearance, more predictable border procedures, internationally recognized testing and certification, simpler documentation, stronger logistics reliability, better digital connectivity, and regulatory cooperation with key partners. These margins reduce the effective cost of participating in production networks and are especially important in standards-intensive sectors such as electronics, machinery, transport equipment, electric vehicles, and other high-growth manufacturing activities.

The main conclusion from the results is that the development challenge for many middle-income economies is not captured by a choice between trade openness and productivity reform. The two are mutually reinforcing. Lower trade costs raise the returns

The second priority is domestic capability upgrading. Trade-cost reductions generate larger gains when firms have the productivity and organizational capacity to exploit new opportunities. Policy should therefore support technology adoption, managerial upgrading, supplier development, workforce skills, quality management, digital production systems, and access to finance for upgrading. These interventions are especially important for firms seeking to move from assembly and basic processing toward design, engineering, testing, systems integration, logistics coordination, and after-sales services.

A central part of this agenda is strengthening the services embedded in manufacturing and high-growth sectors. Modern GVCs rely on engineering, logistics, certification, compliance management, software, data management, marketing, and after-sales support. These activities raise domestic value added even when upstream components remain globally sourced. For middle-income economies, upgrading does not necessarily require producing every input domestically. It requires capturing more value from the functions that surround production.

The third priority is to tailor reform strategies to country structure. Economies deeply embedded in manufacturing GVCs, such as Thailand, Malaysia, and Viet Nam, can gain from reducing interoperability frictions and moving into higher-value manufacturing and service functions. Economies with stronger commodity specialization may need to focus on processing, logistics, standards, and links between resource sectors and downstream industries. Labor-intensive exporters may need to combine market access with supplier upgrading, skills, and compliance capacity. The general principle is common, but the binding constraints differ by country.

The fourth priority is to maintain connectivity through regional integration and selective global links. ASEAN remains a natural anchor for reducing trade costs through trade facilitation, regulatory cooperation, logistics coordination, and mutual recognition. At the same time, middle-income economies need access to extra-regional markets, technologies, and production networks. The goal is not diversification for its own sake, but qualified connectivity: maintaining access to multiple trade communities while reducing the fixed costs and regulatory risks associated with operating across them.

# Fertilizing growth: Agricultural inputs and their effects in economic development

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## Abstract

This paper estimates the role of agronomic inputs in cereal yield improvements and the consequences for countries' processes of structural change. The results suggest a clear role for fertilizer, modern seeds and water in boosting yields. We then test for respective empirical links between agricultural yields and economic growth, labor share in agriculture and non-agricultural value added per worker. The identification strategy includes a novel instrumental variable that exploits the unique economic geography of fertilizer production and transport costs to countries' agricultural heartlands. We estimate that a half ton increase in staple yields generates a 14 to 19 percent higher GDP per capita and a 4.6 to 5.6 percentage point lower labor share in agriculture five years later. The results suggest a strong role for agricultural productivity as a driver of structural change.

## Introduction

Agriculture's role in the process of economic growth has framed a central question in development economics for several decades (e.g., Johnston and Mellor, 1961, Schultz, 1968). While arguments differ regarding the specific mechanisms through which agricultural productivity increases might contribute to structural change in the economy, it has long been theorized that advances in agriculture can promote shifts in labor to higher productivity sectors that offer higher real incomes. Empirical work in more recent years has helped inform the conceptual arguments, and underscored the long-term growth and poverty reduction benefits from agriculture, especially for the most extreme forms of poverty (e.g., Gollin et al., 2007, Ravallion and Chen, 2007, de Janvry and Sadoulet, 2010, Christiaensen et al., 2011). Africa's recent years of economic growth seem to follow a regular pattern of structural change away from agriculture to other economic sectors (McMillan and Harttgen, 2014).

At the same time, additional evidence underscores the role of the manufacturing sector in driving structural change and long-term convergence in incomes across countries (McMillan and Rodrik, 2011, Rodrik, 2013). The theoretical ambiguity on the role of agricultural productivity growth in structural change is captured by the formulation of Matsuyama (1992), which shows opposite effects in closed economies and small open economies. These debates and other evidence regarding agriculture's relatively low value added per worker compared to other sectors (e.g. Gollin et al., 2014) have prompted some researchers to narrow the number of developing countries in which agriculture is recommended as a priority sector for investment (Collier and Dercon, 2014). These issues present a first order concern for understanding why some countries have not experienced long-term economic progress and what to do about it. If agriculture can play a central and somewhat predictable role within the poorest countries, then it is a natural candidate for targeted public investment.

The theoretical and empirical literature regarding structural change is vast, yet statistically identifying the causal role of agricultural productivity is challenging. Indicators of structural change tend to trend together in the process of development, impacts on labor force structure are likely to occur after a lag, and the macroeconomic nature of the question is not amenable to micro-style experiments. Our contribution is to focus on the role of agricultural inputs as drivers of higher yields and subsequent economic transformation, using the unique economic geography of fertilizer production in our identification strategy. The paper builds on the insights of Lagakos and Waugh (2013), which highlights the gaps in understanding of cross-country variations in agricultural productivity. A variety of studies have estimated sources of total factor productivity (TFP) in agriculture in the poorest countries, including in sub-Saharan Africa (e.g., Bates and Block, 2013, Block, 2014), but agriculture is such an input-intensive sector that TFP assessments only provide one piece of the overarching crop sector puzzle. Our cross-country analysis also complements Bustos et al. (2016), who look at structural change in Brazilian municipalities with the diffusion of agricultural technology and practices in soy and maize.

Our econometric strategy proceeds in two parts. First, we empirically assess the inputs that contributed to increased productivity in staple agriculture, as proxied by cereal yields per hectare, during the latter decades of the twentieth century. Using cross-country panel data, this forms a macro-level physical production function for yield increases. We confirm that

fertilizer, modern variety seeds, and water are key inputs to yield growth, even with aggregate data, and controlling for other factors such as human capital and land-labor ratios. Second, we construct a novel instrumental variable (IV) to examine the causal link between changes in cereal yields and aggregate economic outcomes, including gross domestic product (GDP) per capita, labor share in agriculture, and non-agricultural value added per worker. The results provide evidence that increases in cereal yields have both direct and indirect positive effects on economy-wide outcomes, and are particularly pertinent when considering economic growth prospects for countries where a majority of the labor force still works in low-productivity agriculture.

Large-scale nitrogen fertilizer production occurs in a limited number of countries around the world, owing partly to the fact that the Haber-Bosch process requires natural gas. Gravity models have demonstrated that bilateral trade declines in distance, and have been applied to study the relationship between trade and other outcomes such as wealth (Frankel and Romer, 1999) or the environment (Frankel and Rose, 2005). These papers study aggregate trade, but a similar logic can be applied to study the supply of an individual product. Given fertilizer's role as an input in agricultural production, analysis of the economic geography of fertilizer follows the findings of Redding and Venables (2004) in how "supplier access" to intermediate goods matters for trade and income per capita (see also Amiti and Cameron, 2007, Francis and Zheng, 2012). Our identification strategy exploits the global distribution of fertilizer production and subsequent transport distance to the agricultural heartland of developing countries as a source of variation in supplier access. Interacting this distance measure with temporal variation in the global fertilizer price generates an instrument for fertilizer use. To our knowledge this is the first application of economic geography towards causally identifying the relationship between agriculture and structural change.

The IV results suggest that for a country like Mali (one standard deviation above the mean distance measure), a 10% negative price shock to global fertilizer prices would increase fertilizer use by approximately 0.8 kg/ha, increase yields by 7 kg/ha (0.007 tons/ha), increase GDP per capita by 0.3%, decrease the labor share in agriculture by 0.1 percentage points over the following five years, and might increase labor productivity in the non-agricultural sector by around 0.2% over the next nine years. Meanwhile, a country with better supplier access like Jamaica (one standard deviation below the mean distance) would experience a 3.7 kg/ha increase in fertilizer use, a 34 kg/ha (0.034 tons/ha) increase in yields, a GDP per capita increase of approximately 1.3%, a decrease in labor share in agriculture by 0.3 percentage points over the next five years, and an increase in non-agricultural labor productivity of around 0.8% over the nine years (a higher annual growth rate of 0.1%).

The next section of the paper motivates the empirical work, drawing from the many contributions in the literature towards understanding structural change. Section 3 presents empirical models both for estimating the physical production function for cereal yields and for estimating the effect of yield increases on economic growth, labor share in agriculture, and non-agricultural value added per worker. Section 4 describes the data, Section 5 presents the results, and Section 6 concludes.

## The green revolution and structural change

The term "green revolution" was coined following the advent of South Asia's rapid increases in cereal yields in the late 1960s, and is typically used to describe the early stage where yields jump from roughly 1 ton per hectare to 2 or more tons per hectare. Some researchers have argued that these green revolutions underpinned later stages of economic growth, and cite Africa's lack of a green revolution as a key reason why the region has not yet experienced greater long-term economic success (e.g., Diao et al., 2010).

Our empirical work draws on the long theoretical tradition on agriculture-driven structural change dating back to Rostow (1960) and Johnston and Mellor (1961). Mathematical formulations are presented by Matsuyama (1992), Laitner (2000), Hansen and Prescott (2002), Gollin et al. (2002), Gollin et al. (2007), Strulik and Weisdorf (2008), and others. In a stylized story of green revolutions, improvements in agricultural technology are achieved through the introduction of improved land management techniques and improved inputs, including germplasm and fertilizer, all of which boost yields and labor productivity (Murgai, 2001, Restuccia et al., 2008). It is important to note the historical complementarity of agronomic inputs. Modern variety seeds were very fertilizer responsive, and success was often predicated on good water management investments and broader policy support for agriculture. Our economic geography lens leads us to focus on fertilizer, since it is the part of the green revolution package most sensitive to transport costs.

This paper's empirical contribution to understanding the complexity of structural change relates the agricultural labor share directly to agricultural productivity, suggesting that satiation in food demand from Engel's law generates labor movement out of agriculture as that sector's productivity increases. If food is relatively non-tradable beyond local markets, then increased staple food production leads to reduced food prices, increased real wages and hence lower poverty. As staple yields jump and basic food needs are met, crop production begins to diversify, including to cash crops (possibly for export), and so the

virtuous cycle of commercial farming begins. With greater savings and access to finance, farms begin to substitute capital for labor, and freed up workers begin to look for wage employment, typically in nearby cities. To the extent that other sectors enjoy higher labor productivity, this is welfare enhancing.

It is also possible (and we test this empirically) that agricultural productivity improvements trigger further increases in non-agricultural labor productivity. One potential mechanism is that after subsistence needs are surpassed, savings rates increase, and the subsequent capital accumulation increases worker productivity (Lewis, 1954). In parallel, governments are able to collect revenues to finance growth-enhancing infrastructure, such as roads and ports, which increases worker productivity in manufacturing and services. Kuznets (1968) summarizes four main channels through which agricultural growth contributes to economic growth: a forward linkage effect (agriculture providing food and raw materials to non-agricultural production), a backward linkage effect (agriculture consuming industrial products such as insecticide or tractors), inter-sectoral transfers (agriculture contributes taxes and cheap labor to other sectors), and foreign exchange (through agricultural exports). Another mechanism may be that increased farmer incomes improve health outcomes, thus increasing worker productivity, decreasing child mortality, reducing total fertility rates, increasing investment per child, and decreasing demographic pressures. Or, it may simply be that the non-agricultural sector enjoys increasing returns to scale due to fixed costs or learning-by-doing,<sup>1</sup> which would imply that a green revolution and the resulting labor shift would accelerate productivity growth in that other sector. Although our paper is not able to pinpoint which of these mechanisms is at work, our contribution is to provide a causal framework for evaluating whether higher staple yields trigger labor shifts away from agriculture and faster growth in non-agricultural labor productivity.

## Motivating evidence

Trends in the data support the link between staple crop yields and economic growth. Fig. 1 presents cereal yields per hectare from 1961–2001.<sup>2</sup> All developing regions except Africa experienced major sustained growth rates in land productivity over the period, despite varying starting points, and all except Africa more than doubled yields by 2001. Fig. 1 Cereal yields across developing regions, 1961–2001. Fig. 1

A simple Boserup (1965) hypothesis would argue that Africa's yield stagnation relative to other regions is a product of its land abundance, and yields will increase as land becomes scarce. There are three main reasons why this hypothesis does not hold (McArthur, 2013). First, the history of 20th century yield take-offs in the developing world was predominantly characterized by proactive public policies supporting a package of yield-boosting inputs, rather than by factor scarcity (Djurfeldt et al., 2005). These policies are thought to explain much of the regional variations in fertilizer use since 1960, as shown in Fig. 2. Second, labor/land ratios vary tremendously across Africa but they are just as high or higher in many African countries than they were in pre-Green Revolution Asian countries. Third, land productivity is driven by the crucial latent variable of soil nutrients, which are being depleted at dramatic rates throughout Africa. This rapid depletion strongly suggests that land pressures in many countries are not being neutralized by extensification. Fig. 2 Fertilizer use in developing regions, 1961–2001. Fig. 2

Fig. A.1 in the Appendix compares the growth of cereal yields to growth in GDP per capita over the 1965 to 2001 period, exhibiting a positive correlation of 0.44. A noteworthy point emerges from comparing 1965 cereal yield levels to subsequent 1965–2001 GDP growth; no country in the sample<sup>3</sup> experienced negative average growth after reaching a yield threshold of 2 tons/ha as of the beginning of the period (see Fig. A.2 in the Appendix). The relationship between cereal yield growth and growth in non-agricultural value added per non-agricultural worker is similar to that in Fig. A.1, suggesting that higher rates of progress in agricultural productivity are structurally correlated with higher growth rates in non-agricultural labor productivity.

## Empirical model

The first part of our empirical strategy establishes a country-level physical production function for cereal yields in order to motivate the emphasis on agronomic inputs in a study of structural change. We introduce our instrument in a first-stage regression for fertilizer use, and show that the fitted value for fertilizer use is a significant determinant of yields in the second stage. The second part of the empirical strategy deploys our instrument (now for yields in the first stage) to identify the causal impact of increased yields on economic outcomes and structural change as measured by GDP per capita, labor shares, and non-agricultural value added per worker.

### Cereal yield production functions

The agricultural production function analysis builds on previous insights dating back to Hayami and Ruttan (1970) and more recent analyses reviewed in Eberhardt and Teal (2013). We apply panel data to identify a cross-country cereal yield production function, using the following baseline fixed effects specification: 
$$y_{it} = \beta_0 + \beta_1 f_{it} + \delta' X_{it} + \eta_t + \epsilon_{it}$$
 where  $y_{it}$  is the average cereal yield per hectare in country  $i$  in year  $t$ ;  $f$  is the average fertilizer use per hectare; and  $X$  is a vector of

controls including precipitation over a calendar year, the share of seeds that are modern varieties, labor inputs, the share of arable land that is irrigated, average years of schooling as a measure of human capital, and physical machinery per hectare. Meanwhile  $\eta t$  is a time period dummy to flexibly capture global trends,  $\mu_i$  is a country fixed effect and  $v_{it}$  is a random error term. The  $\gamma$  superscript indicates a parameter specific to the agricultural production equation, distinct from the economic growth equations below.

The linear approximation strategy is not without limitations. It was chosen over log-linear and log-log approaches since neither of those were found to provide a better fit with the data, and indeed most countries with significant input use have pursued relatively linear fertilizer-yield trajectories, as shown in Fig. 3. This linear relationship is somewhat at odds with the field-level agronomic data that show decreasing returns, but is likely an inherently limited product of the country-level unit of aggregation. We will show that our results are robust to excluding outliers, a potential concern when using levels specifications, and for completeness the robustness section presents results using log specifications instead. Fig. 3 Cereal yields and fertilizer use, selected countries, 1961–2001. Fig. 3

## Economic growth and labor share equations

It is trivial for higher agricultural productivity to be linked to higher economic growth in the same period, since agricultural output is included directly in national accounts. For example, if one holds fixed all prices and production levels in other sectors, a green revolution-style five year doubling of output in a low-income country with 30 percent of GDP in food production would translate mechanically to a 5.4 percent annual real GDP growth rate.<sup>7</sup> For a country with only 15 percent of GDP in food production, the same yield doubling would translate to 2.8 percent annual growth. Of course a major supply expansion would be expected to decrease the price of food and the nominal measured growth rate might be much smaller.

Beyond this mechanical relation between yields and growth, however, the arguments of Sachs et al. (2004) and McArthur and Sachs (forthcoming) posit that increasing agricultural yields in low-income settings creates scope for structural change via increased savings, investment, and TFP as food becomes cheaper and minimum subsistence requirements are met. This hypothesis can be examined a few different ways. First, a cross-country growth equation for GDP per capita captures the mechanical element of agricultural-to-GDP growth plus indirect aspects of increased investment and higher TFP. Second, a cross-country framework can examine the extent to which increases in staple crop productivity trigger labor movement out of agriculture. Third, a cross-country equation for non-agricultural value added per non-agricultural worker can look for signs of more indirect pathways of investment and TFP increase in non-agricultural sectors.

The baseline fixed effect specification is constructed as follows: 
$$(3) \ln(y_{it}) = \rho \ln(y_{i,t-5}) + \lambda_0 + \lambda_1 a_{i,t-5} + \lambda_2 k_{i,t-5} + \lambda_3 r_{i,t-5} + \omega' MAC_{i,t-5} + \eta t + \epsilon_{it} \epsilon_{it} = \mu_i + v_{it}$$

In Eq. (3),  $y$  represents average real GDP per capita in the first set of specifications and non-agricultural value added per worker in subsequent specifications;  $a_{i,t-5}$  is cereal yield per hectare in previous years (the lag structure will be discussed below);  $k_{i,t-5}$  is lagged aggregate physical capital per worker;  $r_{i,t-5}$  is the total fertility rate as a proxy for demographic pressures and capital widening;  $MAC_{i,t-5}$  represents a vector of standard macroeconomic variables used in the growth literature, averaged from years  $t-5$  to  $t$ ; and the  $\gamma$  superscript indicates a parameter specific to the growth equation. The main coefficient of interest is  $\lambda_1$ . Since the regression controls for country-specific effects, growth over the previous period, and initial income per capita within the period, a significant and positive value for  $\lambda_1$  would lend support to the importance of agricultural land productivity in boosting economic growth. As with the yield regression, we use the instrument described above to improve identification of the causal impact of changes in cereal yield. We have earlier established that we have a valid instrument for fertilizer use; by extension it is valid for yields as well. Instrumenting for yields reduces concerns of reverse causality or omitted variables bias.

Note that there is no ex-ante expectation whether  $\lambda_1$  is biased upwards or downwards in the absence of an IV. If, for example, there is an omitted variable such as a pro-rural government policy that distorts markets to help farmers at the expense of the overall economy (such as price floors on agricultural products), then the policy could generate an association between higher yields and slower overall economic growth, biasing  $\lambda_1$  downwards. On the other hand, urban-led economic development leading to higher incomes, higher demand for food, higher food prices, higher profits for farmers, and thus more investment in agronomic inputs and higher yields would indicate reverse causality from economic development indicators to yield growth. This would bias  $\lambda_1$  upwards. Instrumenting for yields also resolves attenuation bias from measurement errors of a coarse variable such as country-level staple yields per hectare.<sup>8</sup>

Fixed effects estimators suffer from dynamic panel bias particularly pertaining to bias on the lagged dependent variable (Wooldridge, 2002, Bond, 2002). As part of our robustness exercises presented in the Appendix, we conduct a complementary estimation strategy for the economic growth equations through the use of Arellano and Bond's (1991) generalized method of moments (GMM) "difference" estimator, which purges the fixed effects.

The specification we employ to study the effect of yield increases on labor share in agriculture follows the same logic as Eq. (3). However, since the share of employment in agriculture is a bounded variable, we do not estimate a dynamic model including a lag of the dependent variable as we do in the GDP or NAVA regressions. The other independent variables, including the instrumented version of cereal yields, remain the same.

## Data

The estimation strategy draws upon a cross-country panel data set constructed for developing countries over the period 1961–2001. Most of the variables are constructed in 5-year intervals over the period from 1965–2000, based on data availability. Descriptive statistics are in Table 1, and a description and source of each variable is listed in Table A.2 in the Appendix. Much of the data comes from the World Bank's World Development Indicators (WDI), including cereal yield per hectare, fertilizer use per hectare, share of agricultural land under irrigation, and tractors per hectare. A new fertilizer measurement protocol was implemented after 2002, so that is the most recent year that can be included in a relevant time series, as reported in WDI 2006. The key cereal yield variable is defined as follows in the WDI: "kilograms per hectare of harvested land, and includes wheat, rice, maize, barley, oats, rye millet, sorghum, buckwheat and mixed grains. Production data on cereal yields relate to crops harvested for dry grain only. Cereal crops harvested for hay or harvested green for food, feed, or silage and those used for grazing are excluded." (World Bank, 2006). The data count double cropping as part of an annual yield measure rather than counting only the yield per harvest.

Table 1 Descriptive statistics for core sample

| All periods                   | 1965–2000            | Variable | Mean                                                | N              | Mean                                                       | N                                                                                                               | Mean            | N                          | Cereal Yield (tons/ha) |
|-------------------------------|----------------------|----------|-----------------------------------------------------|----------------|------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-----------------|----------------------------|------------------------|
| 1.635                         | 881.187              | 32.117   | 5(1.02)                                             | 0(0.60)        | 1(1.34)                                                    | Fertilizer (tons/ha)                                                                                            | 0.059           | 5880.023                   | 730.091                |
| 75(0.10)                      | 0(.040)              | 0(0.14)  | Precipitation (mm)                                  | 12785          | 881276                                                     | 73124                                                                                                           | 275(820)        | 792(809)                   | Modern seeds (%)       |
| 13.255                        | 40.306               | 928.270  | 19(19.5)                                            | 0(0.99)        | 26(26.2)                                                   | Labor-land ratio                                                                                                | 0.565           | 880.497                    | 30.627                 |
| 5(0.72)                       | 0(0.65)              | 0(0.78)  | Irrigation (%)                                      | 13.256         | 810.068                                                    | 15.874                                                                                                          | 17(17.9)        | 15(15.5)                   | 20(20.1)               |
| Years of Schooling            | 3.445                | 442.006  | 85.116                                              | 8(2.24)        | 1(1.47)                                                    | 2(2.36)                                                                                                         | Tractors per ha | 0.005                      | 84870.003              |
| 2720.012                      | 345(0.0090)          | 0(.0051) | 0(0.0192)                                           | GDP per capita | 26405                                                      | 5591924                                                                                                         | 62327           | 773(constant 2005 \$)      | (2546)(1656)(3367)     |
| Investment                    | 19.549               | 915.554  | 21.171                                              | (% of GDP)     | 7(7.9)                                                     | 6(6.1)                                                                                                          | 8(8.5)          | Inflation                  | 60.235                 |
| 9–017.764                     | 387(387.9)           | –(38.9)  | Government Consumption                              | 13.548         | 511.453                                                    | 13.171                                                                                                          | (% of GDP)      | 6(6.2)                     | 6(6.2)                 |
| 6(6.6)                        | Total fertility rate | 5.345    | 886.427                                             | 34.067         | 5(1.65)                                                    | 0(0.98)                                                                                                         | 1(1.60)         | Labor Share in Agriculture | 58.558                 |
| 868.373                       | 49.675               | 23(23.9) | 20(20.0)                                            | 25(25.5)       | Non-agricultural Value Added per worker (constant 2005 \$) | 57054                                                                                                           | 396661335       | 06569(4749)                | 5129(4806)             |
| Global Fertilizer Price Index | 94.758               | 8(36.9)  | Cost-Distance to Nearest Fertilizer Production Site | 5.785          | 88(3.75)                                                   | Note: Values given for sample of 5-year intervals from 1965–2000. Standard deviations indicated in parentheses. |                 |                            |                        |

Human capital is estimated by Barro and Lee's (2013) measure of total years of schooling. Values of real GDP per capita in constant 2005 US dollars are taken from Version 7.1 of the Penn World Tables (Heston et al., 2012). The labor share in agriculture is calculated using agricultural labor force size data from the FAOSTAT database. This is merged with World Bank (2013) data on cereal area planted in order to calculate the labor to land ratio variable. The numerator and denominator are not a perfect match in this instance, particularly when non-food cash crops represent a large share of agriculture, nonetheless the variable proxies for population pressures on land.

The cereal yield production functions include a historical measure of the introduction of green revolution technology from Evenson and Gollin (2003) and previously presented in Conley et al. (2007). The indicator describes modern variety (MV) crops planted as a percentage of all crops planted, weighted by area planted to those crops. The development of modern seed varieties suitable to Africa's unique crop mix and agroecological zones lagged behind the development of high yield varieties relevant to other regions by roughly two decades (Evenson and Gollin, 2003), so this variable captures the highly relevant proliferation of MVs across countries. Data for the variable cover 85 countries from 1960 to 2000, taken in five-year averages.

Monthly gridded precipitation data are taken from the University of Delaware (Matsuura and Willmott, 2012). Values are summed for each year and averaged over the country, and then converted to natural log form. This is an imperfect signal, since it is rain variability during the location-specific crop growing season that matters most, rather than precipitation across an entire year. Constructing such a location-specific precipitation variable focused on local growing seasons is beyond the scope of this paper.

WDI data are used to measure average aggregate investment and government consumption as a share of GDP. Non-agricultural value added is from the WDI and is measured in constant 2005 dollars to net out changes in relative prices. We blend it with FAOSTAT data on non-agricultural labor force to create a measure of non-agricultural value added per worker in constant dollars.

The sample includes only developing countries with available data, since the main drivers of growth in high-income economies are assumed to be innovation and increasing returns to scale in non-agricultural sectors. We use the middle of

sample time period (1985) for country classification. The World Bank income ceiling for developing country status in 2012 was \$12,615, and given that the WDI's GNI per capita data is reported in 2000 U.S. dollars, we deflate the ceiling to \$9,699, and then keep only countries that had below that income in 1985 (keeping all post 1985-observations regardless of their income trajectory). The sample excludes small economies—defined as those with populations of less than 1 million in 1985—and developing economies in Europe, since their agricultural trajectories have been part of the process of temperate latitude technology transfer and were also affected by Soviet-era socialism. Socialist economies are excluded since their mechanism linking agriculture to structural change was likely skewed by central planning, particularly with regards to rural-to-urban migration and linkages otherwise working through price mechanisms in market economies. We exclude IMF-designated fuel exporters (Algeria, Angola, Congo, Iran, Libya, Nigeria, Oman, Trinidad and Tobago and Venezuela) and major diamond producers (Botswana, Guinea and Namibia).<sup>10</sup> Unlike most developing countries these mineral exporters abound in foreign exchange to finance complementary inputs to labor in the non-agriculture sectors, and they are unlikely to have the low aggregate savings rates of countries where much of the population is in low-productivity farming, thus breaking some of the links between agricultural productivity and structural change mentioned earlier. This leaves 75 countries with data on cereal yields and fertilizer, although we limit the sample in the reduced form cereal yield specifications to 69 countries that have data on all variables. In the estimations for economic growth, labor share, and non-agriculture value added we opt for keeping a consistent number of countries that have data for all variables, thus forming a panel of 58 countries (note that some countries lack data during earlier years). The entire sample spans 1965–2000; however, the economic growth, labor share and NAVA estimations include lags which limit the sample period to 1975–2000. The 75-country sample and 58-country subsample are listed in Appendix Table A.3.

## Results

### Cereal yield production functions

Table 2 presents results for fixed-effect regressions with cereal yield per hectare as the dependent variable, covering five-year intervals over the period from 1965–2000. For each representative observation, yields, precipitation, fertilizer, irrigation, tractors and the labor-land ratio are averaged across three years ( $t-1$ ,  $t$ , and  $t+1$ ) in order to focus on structural shifts as opposed to year to year volatility. Column I presents pooled OLS with year dummies. The coefficient on fertilizer is 7.85 and strongly significant, implying that a 1 kg/ha increase in fertilizer is associated with higher yields of nearly 8 kg/ha. In the absence of country fixed effects, we expect this coefficient to be biased upward, since country-specific characteristics such as capital stock and governance are likely to be positively correlated with both fertilizer use and yields. Table 2 Regressions on Cereal Yield. Table 2 Dependent variable: Cereal yield, tons per hectare (at time  $t$ , in 5 year intervals) Pooled OLS Fixed effects estimator Independent variables (I)(II)(III)(IV)(V)(VI)(VII)(VIII) Fertilizer per hectare [t/ha] 7.85\*\*\* 4.54\*\*\* 4.48\*\*\* 3.40\*\* 3.28\*\* 3.11\* 3.28\*\* 3.14\* (2.24)(1.62)(1.60)(1.58)(1.59)(1.61)(1.60)(1.81) ln (Precipitation [mm]) 0.39\*\* 0.39\*\* 0.36\*\* 0.38\*\* 0.37\*\* 0.36\*\* (0.16)(0.16)(0.15)(0.16)(0.15)(0.15) Modern seeds (%) 0.010\*\*\* 0.010\*\*\* 0.010\*\*\* 0.010\*\*\* 0.010\*\*\* (0.003)(0.003)(0.003)(0.003)(0.003) ln (Agricultural Labor/Land Ratio) -0.20 -0.23 -0.19 -0.19 (0.21)(0.21)(0.21)(0.23) Irrigation (%) 0.007 (0.010) Years schooling 0.02 (0.06) Tractors per 100 sq km 3.29 (10.64) N 463 463 463 463 463 463 463 463 (Within) R-squared 0.500 0.620 0.630 0.660 0.660 0.670 0.660 0.66 Countries 69 69 69 69 69 69 69 69 Country Dummies N Y Y Y Y Y Y Y Year Dummies Y Y Y Y Y Y Y Y Notes: Standard errors in parentheses, clustered by country. All variables except schooling and modern seeds are 3 year means measured at 5 year intervals. E.g., “1970” measures means over 1969, 1970 and 1971. The subsequent value averages over 1974, 1975 and 1976. Constant terms, year dummies and country dummies not reported to save space. \*\*\* $p < 0.01$ . \*\* $p < 0.05$ . \* $p < 0.1$ .

Column II introduces country fixed effects and the fertilizer coefficient drops considerably to 4.54. Column III adds (the natural log of) precipitation. The fertilizer coefficient is nearly unchanged at 4.48 and precipitation is significant with a coefficient of 0.39. This coefficient implies that for a country like Uganda with average yields of 1.3 tons (1300 kg) per hectare and average precipitation of 1202 mm (near the 1201 mm sample average), a one standard deviation increase of 78 mm of precipitation would be associated with yield increases of 25 kg per hectare. Though statistically significant, this coefficient likely suffers from attenuation bias due to measurement error in the nationally-averaged precipitation variable. In an unreported regression, we run year-to-year yields on fertilizer and precipitation and find a consistent coefficient of 0.3 on the precipitation variable.

Column IV introduces another critical element of the green revolution package, modern variety seeds, which is significant at the 1 percent level and substantive in magnitude. This is a pure productivity effect. A marginal one percentage point increase in modern seed use is linked to an extra 10 kg per hectare yield, independent of fertilizer. The inclusion of the seed variable results in a slight decline in the fertilizer coefficient to 3.4, substantiating the point that fertilizer-seed packages have complementary effects in boosting yields.

To round out the production function with a measure of labor, column V adds the agricultural labor-land ratio. The variable is insignificant and has no perceptible effect on the other variables. It is worth noting that this table reports results for a consistent set of observations in all estimations, where the limiting variable in terms of data availability is for tractors. When column V is allowed to include all countries with available data (regression not shown), the larger sample results in a stronger association between labor-land ratio and yields, where the coefficient is  $-0.42$  and significant at the 10% level. We opted for presenting a consistent sample across specifications to ease interpretation of coefficients.

Column VI introduces irrigation, the other main source of water for cereal crops. Column VII introduces human capital measured in average total years of schooling. Column VIII introduces the tractors variable to test for the effects of high-cost physical machinery. While these three variables have the expected positive sign, they are not statistically significant in the presence of country and time fixed effects.

## Economic growth and labor share equations

### Robustness checks

A potential concern with the instrument is the endogenous location of fertilizer production sites. We note that most of the sites are in developed countries, and thus not in our sample. The exceptions are the sites located in Argentina, Brazil, China, Egypt, India, Libya and Trinidad & Tobago. If fertilizer plants located endogenously as a result of improved economic outcomes, and if this nearby location triggered agricultural yield increases, then the instrument would be inappropriately coding for the endogenous location dynamics. To assuage this concern, we ran all regressions while dropping these seven countries (around 7–10% of the data depending on the regression). The results from regressions on labor shares remain qualitatively unchanged (the second stage coefficient on yields remains consistent and significant). In the case of non-agricultural value added, the second stage coefficient on yields remains consistent in magnitude, but loses statistical significance. The fact that the coefficients are consistent without exception suggests that the loss of significance is likely due to sample size issues and the results are not being driven by these seven countries. The second point to note regarding endogenous location of fertilizer production is that the Haber-Bosch process for nitrogen fertilizer production requires natural gas, which creates cost advantages to being close to a natural gas field. As mentioned earlier, the correlation between the distance from agricultural centroid to nearest fertilizer production site and the distance from agricultural centroid to nearest natural gas field is 0.76 for the countries in our sample.<sup>16</sup>

Given that we are presenting results in levels, we conduct an outlier analysis to make sure that our results are not being driven by a small number of observations. For both labor shares and non-agricultural value added per worker, we estimate the endogenous equations (yields without the instrument), calculate a Cook's distance, and make sure that our 2SLS results are all robust to excluding the observations with a Cook's distance above 1. Note that if we calculate Cook's distance on the first stage regression using the instrument, no observations are significant outliers. Only two observations in the endogenous labor share regression had a Cook's distance above 1 in the second stage (Benin and Cambodia in the year 2000). Excluding them from the 2SLS specification leads to no change in the point estimates in either the first or second stage. Similarly, when looking at regressions on NAVA per worker, three observations have Cook's distance above 1 in the second stage (Benin, Cambodia and Mongolia in 2000). Dropping these observations from 2SLS leaves the point estimates unchanged.

For completeness, we also examine the GDP and labor share regressions using yields in logs instead of levels, and present the results in Table 8. Jordan has large residuals in the relationship between log yields and the dependent variables, so we present regressions both including and excluding Jordan. All first-stage regressions have F-statistics on the instrument below 10, suggesting that the instrument is weak in these log specifications.<sup>17</sup> In the GDP regressions, log yields have a positive coefficient on the second stage, but only weakly significant (when Jordan is excluded). The magnitude of 3.34 indicates that increasing yields by 0.5 tons/ha (31% above the sample mean yield of 1.6 tons/ha) would increase GDP per capita by 104%. This effect is larger than when analyzing yields in levels. In the regressions on labor shares, the instrument in the first stage is significant to the 10% level and the F-statistic is 3.56. The second stage coefficient is  $-37.71$ , suggesting that increasing yields by 0.5 tons/ha would decrease the labor share by 11.7 percentage points. Excluding Jordan in regressions VII and VIII leads to significance at 5% levels in both the first and second stages. Since the main regressions with yields in levels produce more conservative and more precise estimates, we opt for the levels regressions (with appropriate outlier tests) as the preferred specifications. As mentioned earlier, it is likely that the weaker first stage relationship when using yields in logs results from farmers in low productivity settings being more sensitive to input prices, such that a change in fertilizer prices does not lead to the same proportional change in yields across high- and low-productivity settings.

Table 8 Regressions with yields in logs.

| Dependent variable    | (I)     | (II)                | (III)                    | (IV)   | (V)                 | (VI)       | (VII)                             | (VIII)                            |
|-----------------------|---------|---------------------|--------------------------|--------|---------------------|------------|-----------------------------------|-----------------------------------|
| ln (Yield)            | (t-1)   | ln (GDP pc)         | ln (Yield)               | (t-1)  | ln (GDP pc)         | ln (Yield) | (t-5)                             | Labor Share in Agr.               |
| ln (Yield)            | (t-5)   | Labor Share in Agr. | (I)                      | (II)   | (III)               | (IV)       | (V)                               | (VI)                              |
| Independent Variables | 2SLS    | 2SLS                | 2SLS                     | 2SLS   | 2SLS                | 2SLS       | 2SLS                              | Global Fert Price/Cost-Adj. Dist. |
| to                    | -0.0012 | -0.0009             | Nitrogen Production Site | (t-1)  | (0.0008)            | (0.0006)   | ln (Yield)                        | (t-1)                             |
|                       | 2.483   | 34*                 | (1.57)                   | (1.88) | ln (GDP per capita) | (t-5)      | 0.070                             | 64***                             |
|                       | 0.070   | 57*                 | (0.07)                   | (0.24) | (0.06)              | (0.31)     | Global Fert Price/Cost-Adj. Dist. |                                   |

to-0.0013\*-0.0011\*\*Nitrogen Production Site (t-5)(0.0007)(0.0005)ln (Yield) (t-5)-37.71\*-50.24\*\*(21.12)(21.54)  
N269264269264Countries58575857Within R-squared0.480.580.500.510.540.190.550.77F Test on First  
Stage2.042.463.564.51 CountryDummiesYYYYYYYYYear DummiesYYYYYYYYStandard errors in parentheses, clustered  
by country in both first and second stages.All variables are 3 year means measured at 5 year intervals. E.g., "1970"  
measures means over 1969, 1970 and 1971. The subsequent value averages over 1974, 1975 and 1976.Constant terms and  
country and time dummies not reported to save space. Regressions III/IV and VII/VIII exclude Jordan. \*p<0.1. \*\*\*p<0.01.  
\*\*p<0.05.

We also conduct the following tests: adding region- or country-specific linear trends to the regressions, considering  
agricultural value added instead of yields, and running the dynamic panel regressions using GMM instrumentation. Secular  
trends in our yield, labor share, and labor productivity variables as economic development proceeds might result in spurious  
correlations, or introduce omitted variable bias to our specifications given that development involves many economic  
characteristics changing together. To that end, Table 9 adds region-specific and country-specific linear trends to the IV  
regressions on labor shares and non-agricultural value added using geographic regions as defined by the World Bank.  
Columns I-II add regional linear trends to the labor share specification. The F-statistic on the instrument is 12.44, suggesting  
that the instrument is strong despite partialing out regional linear trends alongside country and year fixed effects. The second  
stage shows that the coefficient on instrumented lagged cereal yields remains significant and consistent in magnitude to our  
core results. Regressions III and IV employ country-level linear time trends instead of region-level trends, which significantly  
decreases the degrees of freedom in the model. The instrument is significant only to the 10% level in the first stage, and has  
an F-statistic of 3.16. The weak first stage leads to an insignificant coefficient on yields in the second stage, although the  
magnitude is larger than that of our core specifications and the sign continues to be negative. Regressions V-VI and VII-VII  
look at NAVA per worker, using region- and country-level trends, respectively. As with labor shares, the first stage regression  
is robust to including region-level linear trends, with the coefficient on the instrument remaining consistent, strongly  
significant, and the first-stage F-test having a value of 21.81. The second stage coefficient on yield is significant to the 10%  
level and consistent in magnitude with the results in Table 7 (0.29 compared to 0.23). Adding country-level linear trends  
significantly reduces the strength of the first stage (F-statistic decreases to 3.37), consequently precision is lost in the second  
stage estimate. Nevertheless, the coefficient remains consistent at 0.28. Our core results are robust to including regional  
linear trends. Adding country-level trends is the strongest statistical test, and although estimates become imprecise, we  
interpret the consistency of the results as supporting the overall findings.Table 9Adding region and country linear  
trends.Table 9Dependent variableLabor Share in AgricultureLabor Share in Agricultureln (non agriculture valueIn (non  
agriculture valueYield (t-5)Yield (t-5)Yield (t-9)added per worker)Yield (t-9)added per  
worker)(I)(II)(III)(IV)(V)(VI)(VII)(VIII)Independent variables2SLS2SLS2SLS2SLS5-year lag in Global Fert  
Price/Cost-j-0.004\*\*\*-0.0013\*Adj. Dist. to Nitrogen Production Site(0.001)(0.0007)5-year lag yield-8.52\*-16.12(3.53)(9.97)  
9-year lag in Global Fert Price/Cost-0.006\*\*\*-0.0030\*Adj. Dist. to Nitrogen Production Site(0.0014)(0.0017)9-year lag  
ln(non ag value per worker)0.26\*\*0.73\*\*\*-0.150.39\*\*(0.10)(0.10)(0.17)(0.16) 9-year lag yield0.29\*0.28(0.15)(0.47)  
N269269264264Countries58585858Kleibergen Paap F Test on First Stage12.443.1621.813.37 Country  
DummiesYYYYYYYYYear DummiesYYYYYYYYLinear

TrendsRegionRegionCountryCountryRegionRegionCountryCountryNotes: Standard errors in parentheses, clustered by  
country in both first and second stages.All variables are 3 year means measured at 5 year intervals. E.g., "1970" measures  
means over 1969, 1970 and 1971. The subsequent value averages over 1974, 1975 and 1976.Constant terms, year  
dummies, and country dummies not reported to save space. Regions include East Asia & Pacific, Latin America & Caribbean,  
Middle East & North Africa, South Asia, and sub-Saharan Africa. \*p<0.1. \*\*\*p<0.01. \*\*p<0.05.

While our analysis has studied the impact of agronomic yield increases on structural transformation, an alternative measure  
is to focus on labor productivity in agriculture. In the Appendix we consider agricultural value added per worker in agriculture,  
with the caveat that this represents labor productivity across the entire sector (including livestock and cash crops), and thus a  
measure less relevant for studying the impact of raising productivity in the staple food sector. In any case, the results  
(presented in Appendix Table A.4) remain consistent in the case of fertilizer raising agricultural labor productivity, and in turn  
raising GDP and reducing labor shares in agriculture. Finally, Appendix Table A.5 presents a NAVA growth framework using  
GMM instrumentation and finds similar agricultural productivity effects on value added in non-agricultural sectors.

## Discussion and conclusion

Our analysis documents the strong links between agronomic inputs – fertilizer, water and modern seeds – and cereal yields  
per hectare, even after a variety of controls are introduced. We employ a combination of fixed effect and instrumental variable  
specifications to posit a causal economy-wide link between, first, input use and yields, and second, yields and various  
measures of economic growth and structural change. We construct a novel instrument exploiting the economic geography of  
fertilizer production, which together with global fertilizer price fluctuations allow for a statistically causal framework. The

cross-country substantiation of both agricultural yield production functions and their links to various dimensions of economic growth and structural change are empirically consequential. Taking the coefficients from Table 3, a representative country with yields of 1500 kg/ha (1.5 t/ha) that introduces an input package to jump from, say, 15 kg/ha to 65 kg/ha of fertilizer use would be expected to see an average yield jump of 147–462 kg/ha (0.147–0.462 tons/ha); while increasing from 10 to 50 percent use of modern seed would be expected to increase yields by up to 480 kg/ha (0.48 tons/ha).

With regards to economic growth and structural change, the IV results suggest that boosting yields from 1,500 to 2,000 kg/ha (1.5 t/ha to 2.0 t/ha) is linked to a range of 14 to 19 percent increase in income per capita, and a 4.6 to 5.6 percentage point lower share of labor in agriculture five years later. There is also suggestive evidence that this yield boost is associated with approximately 9 to 12 percent higher non-agricultural labor productivity after roughly one decade. The estimated effects are identified based on exogenous variation in fertilizer prices, and are robust to the inclusion of controls for investment and standard macroeconomic policy indicator variables. The results suggest that land productivity promotes growth both by supporting changing labor shares and by increasing total factor productivity. Regressions focused on marginal effects of individual variables are not intended to evaluate nonlinear outcomes guided by Leontief-style agricultural production functions and discontinuous policy functions, so the results might underestimate the potential effects of yields.

The evidence in this paper points to strong potential yield and growth effects resulting from policy efforts to support adoption of a green revolution-type package of complementary inputs in economies with low agricultural productivity and a large share of the labor force still in agriculture. As suggested by theory, these effects are not evident in countries that already export large proportions of their agricultural production, usefully limiting for policy-makers the number of countries where support of agricultural production is likely to lead to structural transformation.

The results suggest a particularly strong role for fertilizer, which is highly consistent with field station agronomic evidence. Fertilizer's high private return on experimental plots and in the field suggests some sort of market failure. Scholars debate whether this is due to credit constraints or non-rational behavior on the part of farmers (Duflo et al., 2008, Duflo et al., 2011). Regardless, the evidence presented in this paper suggests social returns from fertilizer use that exceed the immediate private returns, furthering the case for policy efforts.

It is worth briefly describing the main concerns about increasing fertilizer use. One set is environmental. These are legitimate and require foresight in policy planning (Pingali, 2012). However, countries should not simply avoid fertilizers for environmental reasons, since soil degradation induced by fertilizer omission can pose much greater risks to agricultural production (Palm et al., 2004). A second class of concerns focuses on inequality and the potential scale bias of modern inputs. Hayami and Ruttan (1985) review the evidence on the alleged scale bias in the Asian green revolution and find that the evidence does not support that hypothesis. A third set of concerns focuses on both the challenges of governments implementing input support programs and also the challenges of exiting from them in due course. Though there is evidence that subsidy programs can be successful (Dorward and Chirwa, 2011), there is also evidence that they can be subject to elite capture, and concern that their fiscal drag effects can outlive their usefulness (e.g., Pan and Christiaensen, 2012, Pauw and Thurlow, 2014).

While our results provide some evidence for a causal link from agricultural productivity increases to structural change and higher non-agricultural labor productivity, we can only speculate on the mechanisms through which these effects play out. Nevertheless, our identification of a causal link from yield increases to labor composition shifts rules out models where structural change is driven solely by “pull” forces from growing non-agricultural sectors. To the extent that yield increases contribute to increases in non-agricultural labor productivity growth, this suggests that structural change involves more than just the satiation of food needs and the movement of labor into other sectors. This labor share shift somehow accelerates labor productivity growth. One possible channel might be increasing returns in the non-agricultural sector, perhaps through learning-by-doing. Perhaps increased food production lowers average prices and frees up consumers' resources for other consumption and for productive public and private investments, raising labor productivity elsewhere. Or perhaps higher availability of staple foods promotes health and labor productivity across sectors. Identifying more precise causal pathways between staple yields and structural change forms an important topic for future work.